PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

Design and Development of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo

A Capstone Project
Presented to
the College of Information Technology Education
PHINMA University of Iloilo

In Partial Fulfillment
of the Requirements for the Degree of
Bachelor of Science in InformationTechnology

By:

DELA CRUZ, TYRONE JED
BORRES, JOHN DEXTER
DEOCAMPO, MYCA JADE
JANDOQUILE, KC LYN
PORRAS, EARL JABEZ

Data Informatics Track

Second Semester
2025–2026

TABLE OF CONTENTS

CHAPTER 1 - INTRODUCTION----------5

1.1 Background of the Study----------5

1.2 Statement of the Problem----------12

1.3 Objectives of the Study ----------15

1.3.1 General Objective----------15

1.3.2 Specific Objectives----------15

1.4 Scope and Delimitation----------17

1.5 Definition of Terms----------19

CHAPTER 2 - REVIEW OF RELATED LITERATURE----------23

2.1 Foreign Studies----------23

2.2 Local Studies----------26

2.3 Related Literature---------- 28

2.4 Synthesis of the Reviewed Literature----------31

CHAPTER 3 - METHODOLOGY----------38

3.1 Research Design----------38

3.2 System Development Methodology---------- 41

3.3 System Architecture----------47

3.4 Tools and Technologies Used----------54

3.5 Data Gathering Techniques---------- 57

3.6 Testing Procedures---------- 59

3.7 Evaluation Criteria----------62

Agile Methodology----------47

Data Flow Diagram (DFD - Level 0)----------62

Entity Relationship Diagram (ERD)----------64

Use Case Diagram ----------65

CHAPTER 4 - SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION,RESULTS, AND DISCUSSION ---------- 68

4.1 Introduction----------68

4.2 Overview of the Developed System----------69

4.3 System Development Methodology----------71

4.4 System Architecture----------74

4.5 Hardware and Software Requirement----------77

4.5.1 Hardware Requirements----------77

4.5.2 Software Requirements----------78

4.6 Database Design----------79

4.7 Module Descriptions----------85

4.7.1 Administrator Module----------85

4.7.2 Branch Manager Module----------86

4.7.3 Cashier Module ----------86

4.7.4 Inventory Module----------86

CHAPTER 5 - Conclusions and Recommendations----------88

5.1 Introduction----------88

5.2 Summary of Findings ----------88

5.3 Conclusions----------90

5.4 Recommendations----------91

5.5 Impact of the Study----------93

5.6 Summary----------95

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CHAPTER I

INTRODUCTION

1.1 Background of the Study

Luna's Arrozcaldo is a long-established food business in Iloilo City that has served the community since 1967 [1]. As the business grew and operated across several branches, its daily work became more difficult to manage using manual records. Sales, inventory information, customer details, promotions, and branch performance still need to be organized carefully so that management can make timely decisions.

The existing manual process depends heavily on handwritten records and separate branch information [2]. Sales transactions are recorded and summarized by staff, while management may need to collect information from branches before obtaining a complete picture of business performance. This can take time and may lead to arithmetic mistakes, missing entries, delayed reports, and difficulty comparing branch results.

Inventory is an important concern for a food business because products and ingredients must be available when customers need them. When stock information is not updated quickly, a branch may discover a shortage only after service is affected. On the other hand, ordering too much of a perishable item may lead to unnecessary waste. A digital system can help by connecting sales transactions with inventory information and making current records easier to monitor.

Another need is the proper management of customer information. Luna's Arrozcaldo has built customer relationships through food quality and service, but a centralized digital record can make it easier to maintain customer profiles, purchase history, and loyalty-related information. Promotions can also be recorded and monitored in a more organized way.

The study therefore focuses on a web-based Digital Sales Processing System with Centralized Sales Management and Analytical Reporting. The system brings important sales and operational information into one controlled platform. It provides a digital point-of-sale process, centralized

transaction records, inventory monitoring, customer information management, promotion support, and reports that turn recorded sales into useful business information.

The system is intended to support the business rather than change the identity of Luna's Arrozcaldo. Technology is used mainly to make internal sales and management work more organized, while the business can continue its established service practices. This direction is consistent with the documented project goal of using digital tools to improve efficiency, accuracy, and management visibility while respecting the character of a heritage food business.

Another important concern in the daily operation of a food business is the proper recording and organization of sales transactions. In a manual process, employees may need to write down transaction details, calculate totals, and prepare summaries at the end of a working period. When many transactions are handled within a day, these activities can become time-consuming and difficult to review. A digital sales processing system can provide a more consistent way of recording each transaction and storing the information for later use.

For Luna's Arrozcaldo, sales information is important because it provides a clear view of how the business is performing. Sales records can show the products purchased by customers, the amount of each transaction, the payment method used, and the date of the sale. When these records are properly organized, management can review them more easily and use them as a basis for monitoring daily operations. The proposed system therefore gives importance to the accurate recording of sales information as one of its main functions.

Centralized sales management is also important because Luna's Arrozcaldo operates through multiple branches in the Iloilo City area. When information is kept separately at each branch, management may need to gather and compare records before obtaining an overall view of sales performance. This process can require additional time and may make it harder to identify differences between branches. A centralized system provides a common place where authorized users can access and manage relevant sales information.

The centralized approach does not mean that every employee should have the same access to information. Different users have different responsibilities in the business. A cashier mainly needs to process customer transactions, while a branch manager may need to review branch sales and inventory information. Management may require a broader view of the business, including information from different branches. For this reason, the proposed system is designed around user roles so that system functions can be provided according to the responsibilities of each user.

The use of a digital Point-of-Sale process can also help make sales transactions more organized. Instead of depending mainly on handwritten records and manual calculations, transaction information can be entered directly into the system. The recorded information can then become part of the centralized sales records. This creates a connected flow of information from the actual sale to the stored transaction record and eventually to the reports used by management.

This connection is important because sales information is not useful only at the time a customer makes a purchase. The same transaction can contribute to other business records. For example, a completed sale can be associated with the customer, the products included in the transaction, the quantity sold, and the total amount. The information can also contribute to sales summaries and analytical reports. In this way, one properly recorded transaction can support several business activities.

Inventory monitoring is another area that can benefit from a digital sales process. Food businesses need to know which products and items are available so that customer orders can be served properly. When stock information is recorded manually, updating the quantity after every transaction can be difficult. A digital system can connect sales transactions with product and stock information, making it easier for authorized users to monitor available quantities.

The importance of inventory information becomes more noticeable when products have different levels of demand. Some products may be purchased more frequently than others, while certain items may have lower demand. Without organized records, it may be difficult to determine which products need closer monitoring. By keeping product and sales information together, the proposed system can provide management with a more organized source of information for reviewing product activity.

The system also considers the relationship between sales and customer information. Customers are an important part of any food business because repeated purchases can contribute to long-term business relationships. Keeping customer information in an organized digital record can make it easier to review relevant customer details and purchase history. This can also provide a better basis for managing customer-related activities within the business.

Promotions are another part of the proposed system because promotional activities may be connected to customer purchases and sales transactions. When promotions are handled separately from sales records, it may be difficult to review their effect on transactions. By providing a more organized way of recording promotional information, the system can help management maintain clearer records of promotional activities and their related transactions.

The proposed system also supports the preparation of analytical reports. A business may have a large amount of sales information, but the information becomes more useful when it can be summarized and presented clearly. Analytical reporting allows stored transaction data to be viewed in a more understandable form. Instead of reviewing individual transactions one by one, management can use summarized information to examine sales performance and other relevant business activities.

Analytical reports can help management compare sales information across different periods or branches. Such comparisons can make it easier to recognize changes in business performance. For example, management may review sales records to determine which branch or product has

stronger sales activity during a particular period. These observations can support planning and decision-making without requiring management to manually combine separate records.

The system is therefore not limited to replacing paper records with digital records. Its purpose is to connect sales processing, centralized management, inventory information, customer information, promotions, and analytical reporting within one organized platform. This connection is one of the main features of the study because the different activities are related to one another in the actual operation of the business.

Another benefit of having centralized information is improved access to business records. In a manual setup, management may need to request or collect records from different branches before reviewing overall performance. A centralized digital system can make relevant information available through the system to authorized users. This can reduce the need to depend on separate physical records when reviewing basic sales and operational information.

The system can also help reduce the amount of repetitive work involved in preparing reports. When information is already recorded digitally, it can be organized and summarized through the system. This provides a more direct process from transaction recording to report generation. As a result, employees can focus more on their assigned business responsibilities instead of spending excessive time organizing handwritten records.

Accuracy is another reason for developing the proposed system. Manual calculations and repeated copying of information can create opportunities for errors. A digital transaction process can help standardize the way sales information is entered and calculated. Although correct data entry remains important, using a structured system can reduce unnecessary manual computation and make transaction records easier to review.

The availability of organized information can also improve management visibility. Management does not only need to know the total amount of sales; it may also need to understand where the sales came from, what products were involved, and how transactions changed over time. The analytical reporting component of the system is intended to organize these records into useful information that can support a clearer understanding of business operations.

The development of the system is also based on the actual needs of Luna's Arrozcaldo. The goal is not to introduce technology simply because it is available. Instead, the system is designed to address specific difficulties associated with manual sales processing and separate branch records. This approach helps ensure that the proposed features have a clear connection to the activities performed by the business.

The study also recognizes that technology should support the existing identity and operations of Luna's Arrozcaldo. The business has its own established way of serving customers and managing its food operations. The proposed digital system is focused mainly on improving

internal sales processing and information management. It does not aim to remove the personal service and business practices that are important to the establishment.

The centralized database structure also plays an important role in the proposed system. Sales transactions, users, branches, customers, products, categories, and transaction details need to be stored in an organized manner. A structured database allows related information to be connected while reducing unnecessary duplication of records. This provides a foundation for retrieving information when it is needed for daily operations and reporting.

The relationship between transactions and transaction details is especially important. A transaction represents a customer's purchase, while the transaction details identify the products and quantities included in that purchase. Separating these related records allows the system to maintain organized information while still connecting the complete transaction. This structure supports both sales processing and later analysis of sales data.

The system also includes user and branch information because sales activities are performed by authorized users within specific business locations. Connecting users with their assigned branches allows transactions to be associated with the appropriate branch and user. This can help maintain clearer records of where and by whom a transaction was processed.

Customer records are also connected to transactions so that purchases can be associated with the appropriate customer information when applicable. This provides a basis for maintaining customer-related records within the same system. Product records are similarly connected with transaction details, allowing the system to identify the products included in sales and their corresponding quantities and prices.

These relationships provide a foundation for centralized sales management because the information is not stored as unrelated records. Instead, different parts of the business information are connected through the database. This makes it possible for the system to retrieve related information when generating reports or reviewing transactions.

The proposed analytical reporting component uses the stored information to provide management with a clearer view of business activity. Rather than requiring management to manually inspect numerous transaction records, the system can present summarized results based on the available sales data. The reports are intended to support practical questions such as how sales are performing, which products are active, and how branches compare based on recorded transactions.

The results of the developed system support the need for a more organized digital process for Luna's Arrozcaldo. The system brings together the main information needed for sales processing and management in one platform. Its functions are connected to the identified operational

concerns, particularly the need for organized transaction records, centralized information, inventory monitoring, customer management, and analytical reporting.

The development of the system also provides an opportunity to apply information technology concepts to an actual local business setting. Concepts such as database management, web-based development, sales processing, centralized data management, and reporting are applied to a practical business situation. This allows the study to demonstrate how a system can be designed around actual user and operational requirements.

The system's usefulness also depends on the quality of the information entered by its users. A digital system can organize and process information effectively, but inaccurate or incomplete input can still affect the resulting records and reports. For this reason, proper data entry and consistent use of the system remain important parts of successful implementation. The system provides the structure, while users remain responsible for entering and maintaining correct information.

Data privacy is also an important consideration when managing customer and user information. The system may contain personal and business-related records that should only be accessed by authorized users. The study therefore considers appropriate access control and responsible handling of stored information as part of the system's operation. These controls help protect information while allowing users to perform their assigned functions.

The proposed system is also intended to provide a more reliable basis for future improvements. As the business continues to use the system, the accumulated information may provide a useful record of sales activities and operational patterns. This can help identify areas where additional improvements may be needed. The system can therefore serve not only as a tool for current operations but also as a foundation for future digital improvements.

Overall, the development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting responds to the need for a more organized way of handling sales-related information at Luna's Arrozcaldo. The existing manual approach creates difficulties in recording, organizing, consolidating, and reviewing information across branches. The proposed system addresses these concerns by providing a connected digital process for transactions, products, inventory, customers, users, branches, and reports.

The study therefore highlights the practical value of using information technology to support a local food business. Through digital sales processing, centralized management, and analytical reporting, Luna's Arrozcaldo can have a more organized source of operational information. The developed system is intended to support better record management and provide management with clearer information while allowing the business to continue its established service and operations.

In this way, the study demonstrates that digital transformation does not always require changing the character of an established business. A properly designed system can work alongside existing business practices by improving the way information is recorded, stored, accessed, and reviewed. For Luna's Arrozcaldo, the proposed system provides a practical digital foundation for managing sales information and supporting business operations in the Iloilo City area.

A sales information system is useful when it responds to the actual work of the business rather than simply transferring paper forms to a screen. In this study, the central operational concern is the relationship among transactions, branch records, inventory, customer information, promotions, and management reporting. These activities generate connected information. When each activity is recorded separately, staff may spend time reconciling totals and managers may have difficulty confirming whether summaries represent the same period and branch.

A centralized platform can reduce unnecessary duplication by giving authorized users a consistent location for storing and retrieving records. Centralization does not automatically guarantee accuracy: data quality still depends on clear field definitions, appropriate validation, staff training, and controls against unauthorized edits. For example, transaction dates should follow one agreed format, product identifiers should be consistent, and a completed sale should have a traceable transaction identifier. Such rules make records easier to compare and investigate.

The screenshots supplied with the project show a web interface containing a dashboard, a new-order/POS screen, and navigation for areas including inventory, sales reports, analytics, customers, and settings. The dashboard screenshot displays summary cards and charts, while the POS screenshot shows product selection and a current cart. These images support discussion of the interface that was captured; they do not, by themselves, establish live database behavior, security effectiveness, or successful performance under concurrent use. Those claims require separate tests and recorded evidence.

The study should therefore be interpreted as an applied information-systems project. Its contribution is the design and development of a system intended to make sales-related information more organized and available for operational review. The value of the platform should be assessed through defined tasks, data checks, user feedback, and repeatable test cases, rather than through visual appearance alone. This distinction provides a transparent basis for discussing both the strengths of the current implementation and the improvements that may still be needed.

A branch-based food business needs to maintain a shared understanding of products, prices, transaction dates, and stock movements. Even when branches offer similar products, their sales volumes and stock needs may differ. A centralized record can help management compare branches, provided that each transaction preserves the correct branch identifier and the report

applies a consistent date range. Without those controls, a consolidated dashboard may appear complete while containing omissions or double counting.

A reliable sales workflow begins with clear transaction states. A sale may be initiated, reviewed, paid, completed, voided, or refunded, depending on the rules adopted by the business. The system should define which states affect revenue reports and inventory. For example, a cancelled order should not remain counted as a completed sale, and a refund should be represented transparently rather than silently deleting the original transaction. These are recommended control principles; the actual behavior must be verified against the deployed code and database.

The reporting purpose also needs careful definition. A dashboard can summarize recorded information, but it cannot independently explain why a branch performed well or poorly. Revenue may change because of customer traffic, product availability, price changes, promotions, holidays, or incomplete records. Reports should therefore be used as evidence for follow-up questions, not as proof of causation. Managers should compare equivalent periods and review operational context before making decisions.

The proposed system is most useful when its records can be traced from the summary back to individual transactions. A report should identify the period covered, the branch or branches included, the measure being summarized, and any exclusions applied. This supports reproducibility because another authorized user can apply the same filters and check whether the same totals are returned. It also supports transparency by making the meaning and limitations of each metric visible.

1.2 Statement of the Problem

The study addresses the operational difficulties experienced in managing sales and related information through manual processes. The main problems identified are the following:

1. Sales transactions and daily summaries require manual recording and computation, which may result in arithmetic errors, incomplete records, or delayed consolidation.

2. Branch sales information is not immediately available in one central view, making it difficult for management to monitor overall performance without collecting information from individual branches.

3. Inventory information may not be updated quickly enough to support timely action on low-stock items, possible shortages, or excess stock.

4. Customer information and purchase history are not organized in a single system that can support customer management and loyalty activities.

5. Promotions and discounts may be difficult to monitor consistently when their records are handled separately from transaction information.

6. Management needs clearer reports and visual summaries to compare sales performance, identify popular products, and observe sales trends.

7. Different users require different levels of access to protect sales, customer, inventory, and administrative information.

8. Manual monitoring of daily sales makes it difficult to keep sales information updated and organized throughout business operations.

9. Recording transactions separately can make it difficult to trace previous sales and review transaction details when needed.

10. The absence of a centralized sales record makes it harder to compare sales performance across different branches or periods.

11. Manual inventory recording can cause delays in recognizing changes in product quantities after sales transactions.

12. Limited access to updated stock information can make it difficult for staff to respond quickly to products that are running low.

13. Preparing sales reports manually requires additional time because information must be collected, checked, and summarized before it can be reviewed.

14. Without organized sales data, management may have difficulty identifying which products are frequently purchased by customers.

15. Manual records make it more difficult to observe sales patterns and changes in business performance over time.

16. Customer information and previous purchases can be difficult to review when records are maintained separately from sales transactions.

17. Managing discounts and promotional records manually can increase the possibility of inconsistent or incomplete transaction information.

18. The lack of an easily accessible dashboard limits management's ability to view important sales information in one place.

19. Manual collection of branch information can delay management's ability to obtain a complete view of business operations.

20. The absence of centralized analytical reports makes it difficult for management to use sales information for timely planning and business decisions.

These problems show the need for a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. The proposed system is intended to reduce dependence on manual sales records by providing organized transaction processing, centralized business information, inventory monitoring, customer records, and analytical reports. Through these features, the system can help make sales information easier to record, monitor, review, and use for daily business management.

The system is intended to address information organization through digital transaction capture, centralized storage, and analytical summaries. However, the system should not be described as eliminating all human error. Users can still enter incorrect quantities, choose the wrong product, or select an incorrect branch. Appropriate validation, confirmation prompts, permissions, and correction procedures reduce these risks, while audit trails help explain later changes.

The research questions should distinguish system functionality from operational impact. Functionality asks whether a user can complete a defined task, such as recording a sale or retrieving a report. Operational impact asks whether the system changes the time, accuracy, or ease of that task under comparable conditions. A working screen is evidence of interface implementation, but it is not sufficient evidence of improved accuracy or efficiency. Those outcomes require task-based observations and recorded measurements.

For a reproducible assessment, each research question should be mapped to an observable indicator. For sales processing, indicators may include completion rate, transaction calculation accuracy, and the number of steps required. For centralized reporting, indicators may include report generation time, consistency between totals and source records, and ability to filter by date or branch. For usability, indicators may include task completion, user-reported difficulty, and the types of errors encountered. The exact measures should be selected before the evaluation begins.

If the researchers compare a manual procedure with the digital system, the comparison should use equivalent test transactions and clearly state the conditions. Differences in sample size, staff experience, network conditions, or task complexity may influence the result. The report should avoid claiming that the system caused an improvement when the evidence only shows that users preferred the interface or completed a small demonstration.

A transparent evaluation must also record issues that remain unresolved. If a feature is incomplete, not available during testing, or cannot be verified from the submitted evidence, the report should state that limitation. Reporting limitations does not weaken the project; it helps readers distinguish demonstrated capability from intended capability and provides a realistic basis for future development.

This study seeks to address a system that connects sales processing, centralized data management, and analytical reporting in one platform. The system developed in this study responds to these needs through a role-based web application supported by a centralized database. The project documentation describes the system as integrating a digital POS, centralized sales management, inventory tracking, customer management, promotional tools, and analytical reporting.

1.3 Objectives of the Study

1.3.1 General Objective

The general objective of the study is to design and develop a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo in Iloilo City to improve the organization, accuracy, accessibility, and usefulness of sales and related business information.

This study is to design and develop a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo in Iloilo City. The system is intended to improve the way sales transactions, branch records, inventory information, customer details, promotions, and business reports are recorded and managed.

This objective combines software construction with information management. It should be interpreted as the project's intended outcome, while the degree to which the system meets the objective must be established through the implementation evidence and evaluation results included in the research.

The study also aims to reduce the difficulties caused by the existing manual process. By replacing handwritten sales records and repeated manual calculations with a digital point-of-sale process, the system can help make transaction recording more organized, consistent, and easier to review.

Another objective is to provide management with a centralized view of sales information. The system is designed to collect and organize records in one database so authorized users can

monitor sales performance, inventory information, customer records, and other important business activities without depending only on separate branch records.

Finally, the study aims to provide useful analytical reports and dashboard information that can support daily business decisions. Through sales summaries, tables, charts, product performance, and branch comparisons, the system can help Luna's Arrozcaldo understand its sales activities more clearly and use its recorded data to improve business operations.

1.3.2 Specific Objectives

Specifically, the study aims to:

1. Develop a digital point-of-sale process that records sales transactions accurately and reduces manual computation.

2. Centralize sales records from the branches into a single database for easier monitoring and management.

3. Provide inventory information that can be updated from sales activities and monitored by authorized users.

4. Provide customer management features for organized customer profiles and purchase history.

5. Support the recording and management of promotions and discounts.

6. Generate daily, weekly, and monthly sales reports, together with branch and product performance summaries.

7. Present analytical information through clear tables, charts, and dashboard views that are easy for management to understand.

8. Apply role-based access so that administrators, branch managers, and cashiers can use only the functions appropriate to their responsibilities.

9. Test the system for functionality, usability, reliability, and efficiency with representative users.

10. Provide a practical digital foundation that Luna's Arrozcaldo can use for more organized sales and management activities.

11. A centralized-management objective can be supported by branch-aware records and role-specific views. An analytical-reporting objective can be supported by documented metrics, filters, and checks that reconcile summary values with the underlying transactions.

12. A sales-processing objective can be supported by a transaction workflow and test cases covering product selection, quantity, total calculation, payment recording, and transaction retrieval.

1.4 Scope and Delimitation

The study covers the design, development, testing, and deployment of a web-based Digital Sales Processing System for Luna's Arrozcaldo in Iloilo City. The system is centered on sales processing and centralized management. It connects the point-of-sale process with a centralized database so that authorized users can access updated information according to their roles.

The main system functions include user account management, role-based access, sales transaction processing, product and category management, inventory-related records, customer management, promotions and discounts, centralized sales monitoring, and analytical reporting. The reporting functions are intended to provide useful summaries such as sales totals, product performance, branch comparisons, and sales trends.

The system uses the documented project technology stack: PHP for backend processing, HTML, CSS, and JavaScript for the web interface, PostgreSQL through Supabase for centralized data storage, Railway for the documented deployment environment, and Visual Studio Code as the main development environment.

The system supports three main user roles: Administrator, Branch Manager, and Cashier. The Administrator has broad system access; the Branch Manager manages information related to the assigned branch; and the Cashier primarily handles sales transactions and the functions required at the point of sale.

The study is limited to internal sales and management activities. It does not include full accounting or bookkeeping, payroll processing, employee attendance, tax computation, or a separate customer-facing online ordering and delivery platform. Direct integration with external digital payment providers and advanced predictive forecasting are also outside the present scope. These may be considered for future enhancement if the business requires them.

The system also depends on internet connectivity because its centralized database is hosted through a cloud-based PostgreSQL service. The study therefore focuses on the functionality and usefulness of the developed system within the intended business environment rather than claiming that it removes every possible operational limitation.

Scope

This study covers the design and development of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo in Iloilo City. The system focuses on making daily sales activities more organized by allowing transactions to be recorded digitally instead of relying mainly on handwritten records. It also brings sales information into one centralized system where authorized users can monitor business activities more easily.

The system includes sales transaction processing, product and inventory records, customer information, promotions and discounts, branch sales monitoring, and analytical reports. It also provides dashboards, tables, and charts that can help management review sales performance, compare branches, identify popular products, and observe sales trends. The system is intended to support the Administrator, Branch Manager, and Cashier in performing their assigned responsibilities more efficiently.

The version support the existence of a dashboard and a new-order/POS interface, and show navigation labels for inventory, sales reports, analytics, customers, and settings. Other functions may be present in the deployed system, but their behavior should be confirmed through direct testing or code review before being described as fully operational. Screenshots alone cannot confirm persistence, permission enforcement, or correct calculations.

The system is web-based and therefore depends on a compatible browser and network access to the hosted application and database. This creates practical advantages for centralized access but also introduces dependencies such as internet availability, hosting uptime, and server configuration. The study should identify these dependencies and avoid implying that the system works offline unless an offline mode has been implemented and tested.

Delimitation

The study is limited to the internal sales and management operations of Luna's Arrozcaldo within the Iloilo City area. It does not attempt to cover every business activity of the restaurant. The project mainly concentrates on recording sales, organizing related information, monitoring inventory, managing customer records, handling promotions, and producing reports that can help management understand the business more clearly.

The system does not include full accounting and bookkeeping, payroll, employee attendance, tax computation, or other functions that are not directly related to the project's main purpose. It also does not cover a separate online food-ordering and delivery platform or advanced predictive forecasting. These features may be considered for future improvements, while the present study remains focused on providing Luna's Arrozcaldo with a practical and manageable system for digital sales processing, centralized management, and analytical reporting.

Accounting integration, supplier purchasing, payroll, delivery-platform integration, automated demand forecasting, or mobile applications. These should be listed only when consistent with the actual system boundary. A feature should not be classified as excluded if it has been implemented; conversely, it should not be presented as part of the system merely because it is a recommended future enhancement.

The study's findings are also limited by the data and participants used during evaluation. If testing uses sample records rather than complete historical business records, the results cannot establish performance under every real operating condition. If only a small number of users participate, the feedback should not be generalized to all employees. Clearly stating these boundaries helps readers understand what can and cannot be concluded from the project.

1.5 Definition of Terms

- **Analytical Reporting –** The process of turning recorded sales information into reports, charts, summaries, and trends that help management understand business performance.

- **Analytical Dashboard** – A visual system interface that presents important sales information through summaries, tables, charts, or other views.

- **Agile Development Model** – An iterative development approach in which the system is built in manageable cycles and improved through regular user feedback.

- **Branch Performance** – The sales and operational results of an individual branch compared with other branches or previous periods

- **Business Process** – A series of activities performed by the business to complete an operation, such as recording sales, managing inventory, or preparing reports.

- **Centralization** – Does not mean unrestricted access. User permissions should be assigned according to job responsibilities, and branch information should remain identifiable in stored records and reports.

- **Centralized Sales Management –** A setup in which sales information from the branches is stored in one shared database for authorized monitoring and management.

- **Centralized Sales Records** – Sales information collected and stored in one shared database instead of being maintained separately by each branch.

- **Centralized Database** – A single database used as the main storage location for the system's sales and related operational information.

- **Customer Management –** The organized recording and maintenance of customer information and purchase history for business service and loyalty activities.

- **Customer Purchase History** – A record of a customer's previous purchases stored in the system for reference and management purposes.

- **Digital Sales Processing –** The use of a computerized system to record, calculate, store, and manage sales transactions.

- **Data Integrity** – Information remains accurate, consistent, and appropriately related as it is created, updated, and retrieved.

- **Database Management** – The organized handling of data stored in the system to support accurate and efficient operations.

- **Data Security –** The protection of stored business information from unauthorized access, misuse, or loss.

- **Inventory Management –** The process of recording and monitoring available products or stock so that the business can respond to shortages and other stock changes.

- **Information Management** – The organized handling of business information so that it can be stored, accessed, updated, and used effectively.

- **Information System** – An organized combination of technology, data, users, and processes used to support business operations and decision-making.

- **Manual Process –** A business procedure that depends on handwritten recording, manual calculations, or separate records

- **Operational Management** – The process of monitoring and organizing daily business activities to support smooth operations and decision-making.

- **Point-of-Sale (POS) –** The part of the system used by cashier staff to record customer purchases and complete sales transactions.

- **Promotional Management** – The recording and handling of promotions, discounts, and other sales offers within the system.

- **Product Performance –** The measurement of how products perform based on their recorded sales activity.

- **Referential Integrity** – Can be supported by primary and foreign key constraints; MySQL's official documentation explains how foreign keys cross-reference related tables and help maintain consistency

- **Reproducibility** – means that another person can repeat a documented test using the stated inputs, steps, environment, and expected outcome.

- **Research Locale** – The specific place or business environment where the study is conducted; in this study, it refers to Luna's Arrozcaldo in Iloilo City.

- **Role-Based Access Control –** A security approach that gives users access to system functions according to their assigned role.

- **Sales Report** – A generated summary of recorded sales used to review business activities for a specific period.

- **Sales Summary** – A simplified presentation of sales information for a specific day, week, month, branch, or product.

- **Sales Trend** – A pattern or change in sales activity observed over a certain period.

- **System Development** – The process of planning, designing, building, testing, and improving the proposed digital sales system.

- **System Testing** – The process of checking whether the developed system functions properly and meets its intended requirements

- **System Implementation** – The process of putting the developed system into actual use within the intended business environment.

- **System Deployment** – The process of making the completed web-based system available in its intended operating environment.

- **Transaction –** A recorded sale that contains information such as the customer, cashier, products, quantities, payment method, and total amount.
- **Transaction Detail –** A record of the individual products, quantities, prices, discounts, and subtotals included in a transaction.
- **Transparency** – Methods, assumptions, limitations, and evidence are reported clearly.
- **User Acceptance Testing –** Testing in which representative users use the system to determine whether it meets practical needs and is easy to use.
- **User Acceptance Testing (UAT)** – Testing performed by representative users to determine whether the system meets their practical needs and is suitable for use.
- **Web-Based System** – A system that users can access through a web browser to perform supported business activities.

REFERENCES

[1] 2010 September 03 et al., "Luna's arroz caldo: An Ilonggo legacy," FLAVOURS OF ILOILO, https://flavoursofiloilo.blogspot.com/2010/09/lunas-arroz-caldo-ilonggo-legacy.html?m=1

[2] https://ijbss.thebrpi.org/journals/Vol_12_No_10_October_2021/2.pdf

[3] P. Parsa and A. Khan, "Restaurant operations and the role of technology in improving transaction processes," IEEE Journal of Restaurant Technology, vol. 12, no. 3, pp. 145–158, 2011.

[4] J. Highsmith and A. Cockburn, "Agile software development: The business of innovation," IEEE Computer, vol. 34, no. 9, pp. 120–127, Sept. 2001.

[5] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," IEEE MIS Quarterly, vol. 13, no. 3, pp. 319–340, 1989.

[6] S. Negash, "Business intelligence," IEEE Communications of the Association for Information Systems, vol. 13, no. 1, 2004.

[7] R. S. Sandhu, E. J. Coyne, H. L. Feinstein, and C. E. Youman, "Role-based access control models," IEEE Computer, vol. 29, no. 2, pp. 38–47, Feb. 1996.

[8] D. Buhalis and R. Leung, "Smart hospitality—Interconnectivity and interoperability towards an ecosystem," IEEE International Journal of Hospitality Management, vol. 71, pp. 41–50, 2018.

[8] G. Vial, “Understanding digital transformation: A review and a research agenda,” IEEE The Journal of Strategic Information Systems, vol. 28, no. 2, pp. 118–144, 2019.

[10] P. C. Verhoef et al., “Digital transformation: A multidisciplinary reflection and research agenda,” IEEE Journal of Business Research, vol. 122, pp. 889–901, 2021.

PHINMA UNIVERSITY OF ILOILO
College of Information Technology Education

CHAPTER II

REVIEW OF RELATED LITERATURE

This chapter reviews studies and literature that support the development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. The discussion follows the simple organization of the provided capstone proposal guide: foreign studies, local studies, related literature, and synthesis.

2.1 Foreign Studies

Villafuerte Bazán et al. (2024) examined the use of business intelligence as a strategy for decision-making in the sales area of SMEs. Their study reported improvements in sales-related indicators and decision-making after the use of a technological BI solution. The study also emphasized the value of timely information and analysis in helping management make faster and more informed decisions. This relates directly to the analytical reporting component of the proposed system, particularly its purpose of presenting sales totals, product performance, branch comparisons, and sales trends.

Ocloo, Coffie, Bukari, and Bashiru (2024) examined the digitization of small and medium-sized restaurant enterprises and studied how owner or manager IT skills and attitudes influence digital adoption. The study involved 261 restaurant SMEs and found that digitalization is affected by the ability of businesses and their managers to adapt to technology. This is relevant to Luna's Arrozcaldo because the proposed system is intended for actual restaurant operations where administrators, branch managers, and employees need to use digital tools as part of their daily work. The study supports the idea that successful system implementation depends not only on the software itself but also on how easily the people in the business can understand and use it.

Bar-Gill, Brynjolfsson, and Hak (2024) investigated how analytics tools can help small businesses become more data-driven. Their field experiment involving eBay sellers found that access to a data-rich dashboard increased data-driven decision-making and was associated with higher seller revenues. The researchers also emphasized the importance of active performance

monitoring when using analytics tools. This supports the proposed Luna's Arrozcaldo system because its analytical reporting and dashboard features are designed to help management monitor sales performance, compare results, and use recorded information when making business decisions.

Orjatsalo, Hussinki, and Stoklasa (2024) examined how managers use business analytics when making decisions. Their findings showed that managers use analytical information to monitor business performance against targets and to identify actions that may be needed. This supports the proposed system's use of analytical dashboards and reports because Luna's Arrozcaldo management can use sales information to monitor branch performance, product performance, and sales trends.

Anuradha and Shri Sailaxmi (2024) studied business analytical tools that can help small and medium-sized enterprises improve their operations. The study examined different industries and highlighted the usefulness of affordable analytics for understanding customer behavior, operational efficiency, and market trends. This supports the proposed system because Luna's Arrozcaldo can use its recorded sales data to understand product performance and business trends without relying only on manual summaries. A (2024) study on data-driven analytical systems for small and medium-sized enterprises examined how businesses can use data-driven methods to improve operational management. The study emphasized that businesses that still rely heavily on traditional processes may face difficulties in using available information effectively. It proposed a data-driven framework that connects business activities with analytics to support operational improvement. This relates to Luna's Arrozcaldo because the proposed system connects sales processing, centralized information, and analytical reporting in one platform.

Ikomu (2024) developed a restaurant management system for small-scale restaurants in Uganda. The study addressed problems associated with manual processes, limited technology adoption, and operational efficiency. The resulting system included functions such as product management, customer relationship management, staff management, reporting, and online ordering, with reported improvements in data management and reporting. Although its setting is different from Luna's Arrozcaldo, the study is relevant because both projects focus on using a digital system to organize restaurant operations and business information.

Ragazou, Passas, Garefalakis, and Zopounidis (2023) examined the role of business intelligence in helping SMEs improve decision-making and competitive advantage. Their study highlighted how BI can help businesses make better use of information and develop capabilities through emerging technologies. This is related to the analytical reporting component of the Luna's Arrozcaldo system because recorded sales information can be converted into useful summaries, trends, and performance information instead of remaining as simple transaction records.

Alsibhawi, Yahaya, and Mohamed (2023) studied business intelligence adoption among small and medium-sized enterprises and developed a framework based on factors that influence the

adoption of BI systems. Their study emphasized the need for SMEs to use information technology to address problems associated with traditional business processes. This relates to the proposed system because Luna's Arrozcaldo also needs a more organized way of handling sales information instead of depending heavily on separate or manual records. The study supports the use of centralized information and business intelligence features to improve the way business data is managed and used.

Wailgum (2013) discussed cloud-based POS systems and their value for smaller businesses. Cloud-based systems can reduce dependence on expensive dedicated hardware and can make business information available through internet-connected devices. This supports the study's use of a web-based system for Luna's Arrozcaldo, where authorized management users need access to centralized information.

Wixom and Watson (2010) discuss the relationship between business intelligence and organizational practice. Applied to this project, the lesson is that technical reporting is only one part of an information system. Managers need shared definitions for measures such as total sales, number of transactions, and product performance. If different users calculate the same measure in different ways, a centralized dashboard may reproduce disagreement rather than resolve it. Metric definitions and report filters therefore belong in the system documentation.

Negash (2004) explained business intelligence as the use of data and analytical tools to support planning and decision-making. This is closely related to the analytical reporting component of the Luna's Arrozcaldo system because raw transaction records become useful summaries, comparisons, and trends for management.

Highsmith and Cockburn (2001) discussed Agile software development and emphasized close cooperation between developers and users. This supports the use of Agile for the study because the system requirements are based on actual business activities. User feedback can be used to improve features during development instead of waiting until the end of the project.

Foreign studies on point-of-sale systems show that digital transaction processing can improve speed, accuracy,and the quality of sales records. Parsa and Khan (2011) discussed restaurant operations and the role of technology in improving transaction processes. Their work supports the use of POS technology to reduce manual calculation and improve order accuracy, which is relevant to a food business that processes many transactions during busy service periods.

Ferraiolo and Kuhn (1992) introduced the role-based access control approach, in which permissions are assigned according to roles. This supports the system's use of different access levels for administrators, branch managers, and cashiers. It helps protect information by limiting users to the functions needed for their work.

Davis (1989), through the Technology Acceptance Model, identified perceived usefulness and perceived ease of use as important factors in users' acceptance of information technology. The study developed and validated measures for these two factors and found that perceived usefulness was strongly associated with technology usage. This supports the need for the proposed sales system to have clear functions and simple workflows because cashiers, branch managers, and administrators must be able to use the system effectively during actual business operations.

2.2 Local Studies

(Gańgan, 2025) presented a web-based market goods trading system for the Agri-Pinoy Trading Center, demonstrating the use of a web application and relational database for transaction recording and data management. Its focus on organized data and digital processing supports the database-centered approach of the proposed Luna’s Arrozcaldo system.

Usita et al. (2025) evaluated iProSES, a web-based inventory management system for food-sector micro, small, and medium enterprises in the MIMAROPA region of the Philippines. The researchers used surveys, system usage logs, and interviews to assess system effectiveness and user acceptance. Their findings reported an overall user satisfaction score of 4.68, with positive ratings for usefulness and ease of use. However, the study also identified internet connectivity and maintenance costs as challenges that may affect implementation. These findings are relevant to Luna’s Arrozcaldo because they emphasize the importance of developing a practical, user-friendly digital system that improves business record management while considering technical limitations and long-term maintenance requirements.

Moreno, Juanitas, Montales, and Bauzon (2025) examined the Challenges and Perception on the Adoption of Point-of-Sale Systems among Micro and Small Enterprises in Tacurong City. Their quantitative study involved 39 respondents and investigated the relationship between POS adoption challenges and business owners’ perceptions of the technology. The results indicated generally positive perceptions, while operational challenges were identified as an important factor influencing acceptance. The researchers also found that business size and the type of POS system were associated with differences in adoption experiences. This study supports the development of Luna’s Arrozcaldo system by emphasizing the need to consider the actual requirements, technical familiarity, and operational conditions of its intended users. A practical and understandable system may help make digital sales processing more manageable for a small food business.

Philippine studies provide a direct context for the proposed system because many local businesses experience challenges involving manual records, limited resources, and the need for practical digital tools. (Barredo et al., 2023) documented Bugana, an electronic marketplace for food producers in Bago City, and showed how digital transaction processing can improve order

handling and reduce reconciliation problems. This supports the use of organized digital transaction recording for food-related businesses such as Luna's Arrozcaldo.

(Briones, Galang, & Latigar, 2023) examined digital transformation in Philippine agri-food systems and highlighted the importance of staff digital skills, technology infrastructure, data management, and implementation support. These findings are relevant to Luna's Arrozcaldo because the effectiveness of a digital sales system depends not only on its functions but also on proper use and staff readiness.

(Barredo et al., 2023) documented Bugana, an electronic marketplace for food producers in Bago City, and showed the practical benefits of digital transaction processing in improving order handling and reducing reconciliation problems. This study is relevant to Luna's Arrozcaldo because organized digital recording can help reduce the difficulties associated with manually managing sales transactions.

(Quimba & Calizo, 2019) examined the adoption of POS technology among Philippine food-service MSMEs and identified cost, perceived complexity, and uncertainty about the return on investment as important barriers to adoption. This finding supports the development of a practical web-based system that focuses on the actual needs of the business instead of unnecessary complexity. For Luna's Arrozcaldo, the proposed system addresses this concern by bringing sales processing, centralized management, and analytical reporting into one platform

(De Jesus, 2017) examined e-commerce adoption among agribusiness MSMEs in Cavite and found that owner-manager digital competence and previous exposure to digital tools were important factors in technology adoption. This finding highlights that business owners and employees need sufficient familiarity and confidence when introducing digital systems into existing operations. It supports the proposed system's emphasis on simple workflows and user-friendly functions that can be applied to the daily activities of Luna's Arrozcaldo.

(Encarnacion & Genodia, 2016) documented an online ordering system for H&J Bakeshop in Iloilo City. The study demonstrated how a local food business can use a digital system to improve ordering while considering ease of use and accuracy. This is relevant to Luna's Arrozcaldo because the proposed system will also be used during actual food-service operations. Philippine studies provide a relevant foundation for the proposed system because local food businesses and MSMEs face challenges in organizing transactions, managing information, and adopting appropriate digital tools.

Chicano and Huervana (2009) developed a Computerized Restaurant Sales and Stock Monitoring System for a specialty restaurant in Iloilo City. Their study identified manual billing and unmonitored inventory as operational problems and proposed a computerized point-of-sale system to improve sales management, stock monitoring, and the protection of encoded information. This research is closely related to Luna's Arrozcaldo because both studies address

the difficulties of manual transaction recording and inventory monitoring in food-service businesses. The findings support the use of a digital system to organize sales transactions, maintain accurate records, and provide better control over daily operations. Source: Central Philippine

Overall, these local studies show that digital systems can be practical for Philippine food and small-business settings when they are designed around actual business workflows. They also support the importance of organized transaction recording, accurate data management, user participation, and analytical information, which are directly connected to the proposed Design and Development of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo.

2.3 Related Literature

MySQL documentation (2026) describes transaction-related statements that allow database operations to be managed as coordinated units of work. Appropriate transaction handling is important when several database changes must remain consistent during a business operation. For Luna's Arrozcaldo, this concept is relevant when recording sales and updating associated records, provided that the implemented database and application logic support the required transaction behavior. This literature supports testing whether failed operations are handled correctly and whether sales information remains consistent after successful or unsuccessful transactions.

The evolution of information systems has transformed the manner in which organizations perform their daily operations. Traditional business processes that relied heavily on paper-based documentation and manual procedures have gradually been replaced by computerized systems capable of processing large volumes of information with greater speed, consistency, and accuracy. This transformation has enabled organizations to reduce operational costs, minimize human error, improve customer service, and strengthen managerial control through real-time access to organizational information.nConsequently, information systems have become strategic organizational 83 assets that support innovation, operational excellence, and sustainable competitive advantage (Laudon & Laudon, 2022).

Restaurant Management Systems (RMS) have evolved from basic electronic billing and Point-of-Sale functions into integrated digital systems. They centralize customer orders, inventory, reservations, and sales records in one database. Automation reduces duplicate entries and human errors while keeping information accurate and updated. These systems also generate real-time sales reports and dashboards, helping managers monitor performance and make better decisions (Prakash et al., 2025).

The World Wide Web Consortium's Web Content Accessibility Guidelines (WCAG, 2024) 2.2 provides recommendations for making web content more accessible to users with different abilities and needs. These guidelines address areas such as perceivability, operability, understandability, and robustness when interacting with digital interfaces. For Luna's Arrozcaldo, applying relevant accessibility principles can help make sales processing and management pages easier to navigate across supported devices and screen sizes. The guidelines provide a useful reference for reviewing interface accessibility, although responsiveness and accessibility must still be verified through actual system testing.

Restaurant Management Systems (RMS) have become indispensable technological solutions within the foodservice industry because they automate and integrate various processes into a centralized information system. As restaurants experience increasing customer demands, diverse service offerings, and complex operational activities, the implementation of computerized management systems has become 87 essential in maintaining efficiency, accuracy, and service quality. Unlike traditional manual procedures that rely heavily on paper-based records, Restaurant Management Systems provide an integrated platform that manages customer orders, Point-of-Sale (POS) transactions, reservation scheduling, inventory monitoring, employee management, kitchen operations, financial reporting, and business analytics within a single application (Laudon & Laudon, 2022).

The National Institute of Standards and Technology (NIST, 2022) published the Secure Software Development Framework to recommend practices for reducing software vulnerabilities throughout the development life cycle. The framework emphasizes integrating security practices into software development instead of treating security as a separate task after implementation. For Luna's Arrozcaldo, these principles are relevant to protecting user accounts, controlling access to sales information, and reducing risks associated with insecure application functions. The framework can guide security reviews and testing, but the researchers must verify the system's actual safeguards before claiming that specific security controls have been implemented successfully.

Another important aspect of digital transformation is the generation and utilization of organizational data. Modern organizations continuously collect operational information from customer transactions, inventory activities, reservation records, employee performance, and financial reports. Through digital transformation, these operational data are converted into valuable business intelligence that supports evidence-based decision-making. Managers utilize analytical tools to identify customer purchasing behavior, evaluate sales performance, forecast demand, monitor inventory consumption, and assess operational efficiency. Consequently, digital transformation improves not only business automation but also organizational learning and strategic planning (Verhoef et al., 2021).

Digital transformation has become one of the most significant developments influencing organizations across various industries, including the hospitality and food service sectors. It refers

to the integration of digital technologies into business processes, organizational strategies, and customer interactions to improve operational efficiency, service quality, and organizational competitiveness. Unlike simple computerization, digital transformation involves redesigning business operations by utilizing emerging technologies such as cloud computing, artificial intelligence, mobile applications, data analytics, and integrated information systems. Through digital transformation, organizations are able to automate routine tasks, improve collaboration, optimize resource utilization, and respond more effectively to changing market conditions (Vial, 2019).

Integrating customer relationship management, inventory control, reservation management, online ordering, business intelligence, cloud computing, and mobile accessibility. These technological developments have enabled restaurants to streamline daily operations, reduce administrative workload, improve communication among departments, and provide customers with faster and more reliable services (Buhalis & Leung, 2018).

Modern Business Process Automation also supports organizational transparency and accountability. Automated systems maintain comprehensive transaction histories, user activity logs, and operational records that allow management to monitor employee performance and organizational activities more effectively. Audit trails generated by automated systems facilitate organizational control by documenting every transaction performed within the system. These capabilities improve data integrity, strengthen internal control mechanisms, and support organizational compliance with established business procedures (Dumas et al., 2018).

Business intelligence literature explains how organizations can use collected business data to support analysis and decision-making. Choudhuri et al. (2011) discussed the importance of intelligent technologies in transforming business data into useful information for organizations. This supports the analytical reporting component of the proposed system, where recorded sales data can be organized and presented to help Luna's Arrozcaldo understand sales activities and business performance.

Negash (2004) explained that business intelligence combines data collection, analytical tools, and information presentation to support organizational decision-making. Business intelligence is relevant to sales management because recorded transactions can be converted into reports that help managers understand business performance. For Luna’s Arrozcaldo, analytical reporting can organize sales information into daily summaries, product performance reports, and other useful indicators supported by the system’s actual data. This literature supports the inclusion of reporting functions that help management review sales activities and make evidence-based decisions.

Database literature also emphasizes the importance of properly organized and related data in information systems. Date (2003) explained that relational database systems organize information through related tables to support accurate storage and retrieval. This is relevant to the proposed

system because centralized sales records, product information, customer information, and transaction details need to be properly organized to support reliable reporting and analysis.

DeLone and McLean (2003) presented an updated information systems success model that explains how system quality, information quality, service quality, system use, user satisfaction, and net benefits relate to information system success. This model highlights that an information system should provide reliable functions and useful information rather than merely operate as a digital replacement for manual work. In Luna's Arrozcaldo, the quality of sales records, the usability of management functions, and the usefulness of generated reports are important considerations when evaluating the proposed system. The model provides a theoretical basis for assessing whether the system supports business operations and satisfies its intended users.

The Technology Acceptance Model of Davis (1989) further explains that perceived usefulness and ease of use influence users' acceptance of information technology. This is relevant to the proposed system because the digital sales processing system should provide practical functions that personnel can understand and use during daily sales operations. These concepts support the development of a system that combines organized data management, sales processing, and analytical reporting in a user-oriented manner.

2.4 Synthesis of the Reviewed Literature

The reviewed studies and literature consistently support the main direction of this capstone. Digital POS systems can improve transaction recording; centralized databases can improve access to branch information; analytical reporting can turn sales records into useful management information; inventory systems can support better stock monitoring; customer management can strengthen loyalty activities; and role-based access can help protect business information.

The local studies are particularly important because they show that digital systems can be applied in Philippine business settings with limited resources. They also show that usability and user participation matter. A system that is too difficult to use may not provide its expected benefits even when its technical features are complete.

The review also identifies a practical gap. Many studies discuss individual functions such as POS, inventory, reporting, or customer management, but fewer focus on an integrated system designed specifically for a multi-branch heritage food business in Iloilo City. The present study addresses this gap by combining these functions in one platform and adapting the system to Luna's Arrozcaldo's actual operational needs.

Overall, the literature supports the study's central idea: a simple, centralized, and user-focused digital system can help a growing food business organize its sales information, improve

management visibility, and make better use of the data it already produces. The source capstone document similarly identifies the lack of integrated systems for regional Philippine heritage food businesses as a research gap and positions the project as an integrated digital solution for Luna's Arrozcaldo.

Business intelligence literature further highlights the value of transforming stored business data into information that can support decision-making. Wixom and Watson (2010) discussed how business intelligence systems help organizations collect, manage, analyze, and present data for improved decision-making. This relates to the analytical reporting component of the proposed system, where recorded sales information can be processed into reports and useful summaries that may help Luna's Arrozcaldo monitor sales performance and recognize purchasing patterns.

Data warehousing and analytical systems also provide a foundation for examining historical business information. Kimball and Ross (2013) emphasized the importance of organizing business data in a way that supports reporting and analysis. This concept is applicable to the proposed system because historical transaction records can provide a basis for examining sales trends, frequently purchased products, and customer purchasing behavior.

The usability of a system is also important because personnel need to interact with the system during actual business operations. Davis (1989) explained through the Technology Acceptance Model that perceived usefulness and perceived ease of use influence users' acceptance of information technology. This supports the need for a practical and understandable interface in the proposed system so that personnel can efficiently perform sales-related tasks while accessing the information needed for daily operations.

The reviewed foreign and local studies, together with the related literature, provide a clear foundation for the development of a digital sales processing system for Luna's Arrozcaldo. The reviewed materials generally emphasize the importance of using information technology to organize business activities, manage information, improve transaction processing, and support decision-making. Although the studies differ in their specific applications and settings, they share the idea that properly designed digital systems can make business processes more organized and information easier to manage. These concepts are directly related to the proposed Design and Development of a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo.

The reviewed literature first establishes the importance of digital systems in improving the way organizations handle information. Digital systems allow information to be recorded, stored, retrieved, and processed through a structured environment. This is particularly useful for businesses that regularly handle transactions and maintain records of their operations. Instead of depending entirely on separate manual records, a digital system can provide a more organized approach to storing information. This supports the proposed system because Luna's Arrozcaldo

needs a practical way to manage sales information while maintaining records that can be used for monitoring and reporting.

The studies also emphasize the value of digital transaction processing. Barredo et al. (2023), as discussed in the local studies, presented an electronic marketplace for food producers in Bago City and highlighted the practical value of digital transaction processing in improving order handling and reducing reconciliation problems. This is relevant to the proposed system because sales transactions are an important part of the daily operations of Luna's Arrozcaldo. The finding supports the idea that transaction activities can benefit from a digital process that organizes recorded information and reduces the difficulties associated with handling transactions manually.

The local literature further demonstrates that digital transformation is applicable to Philippine business environments. Briones, Galang, and Latigar (2023) discussed digital transformation in Philippine agri-food systems and highlighted the importance of leadership support, staff digital skills, technology infrastructure, data management, and implementation support. These points are relevant to the proposed system because successful system implementation does not depend only on the software itself. The personnel who will use the system and the business environment in which it will operate are also important considerations. For Luna's Arrozcaldo, the system needs to be understandable and suitable for actual business operations.

Another local study reviewed in the literature is the work of Gaingan (2025), which presented a web-based market goods trading system for the Agri-Pinoy Trading Center. Its use of a web application and relational database demonstrates the application of digital processing and structured data management in a local setting. This supports the database-centered approach of the proposed Luna's Arrozcaldo system. The connection is important because the proposed system also requires a structured database where sales, products, customers, and transaction information can be stored and retrieved when needed.

Encarnacion and Genodia (2016) also provided a local example through an online ordering system for H&J Bakeshop in Iloilo City. The study is relevant because it demonstrates how a local food business can adopt a digital system while considering ease of use and order accuracy. These concerns are also applicable to Luna's Arrozcaldo, where personnel need to use the system during actual food-service operations. The study therefore supports the importance of developing a system that is not only functional but also practical for its intended users.

The local studies collectively show that digital systems can be useful within Philippine food and small-business environments. Their findings emphasize practical implementation, staff readiness, organized data, and systems that are designed around actual workflows. These ideas support the proposed system because Luna's Arrozcaldo requires a solution that fits its business activities rather than a system that is designed without considering the users and their daily tasks. The reviewed local studies therefore provide contextual support for developing a digital system appropriate to the business environment of the study.

The reviewed literature also establishes the importance of proper database organization. Date (2003) explained the role of relational database systems in organizing information through related structures. This concept is important to the proposed system because different types of business information need to remain connected. Users, customers, products, transactions, and transaction details represent different information but are related through the activities of the business. Organizing these records into appropriate database structures provides a foundation for retrieving information and preparing reports.

The importance of database organization becomes clearer when considering the analytical requirements of the proposed system. Sales transactions contain information that can later be examined to understand business activities. However, the information can only be useful for analysis if it is stored consistently and can be retrieved properly. A structured database therefore supports both operational transactions and analytical reporting. This connects the database design of the proposed system with its broader purpose of providing centralized sales management.

Centralized data management is another major concept supported by the reviewed literature. A centralized system provides a common location for storing important business information. This can make related records easier to access and manage compared with maintaining information across disconnected sources. For Luna's Arrozcaldo, centralized sales management means that relevant sales information can be maintained within the proposed system and used by the appropriate users. This supports a more organized approach to manage the business's sales records.

The reviewed literature also highlights the importance of information quality. DeLone and McLean (2003) emphasized information quality as one of the important dimensions associated with information-system success. This is relevant to the proposed system because the usefulness of reports depends on the quality of the information recorded in the system. If transaction information is properly recorded and organized, it can provide a more reliable basis for reporting. Therefore, accurate data entry and proper data management are important considerations in the proposed digital sales processing system.

The relationship between data quality and analytical reporting is particularly important. Analytical reports are based on the information stored within the system. If the underlying transaction records are incomplete or poorly organized, the resulting reports may not provide a clear representation of business activities. The proposed system therefore connects transaction processing with analytical reporting so that the information generated from sales activities can be organized and presented in a useful manner. This relationship is one of the important foundations of the proposed system.

The reviewed literature also supports the use of historical transaction data for business analysis. Sales records can provide information about previous purchasing activities and can be examined to identify patterns over time. The proposed system uses this concept by maintaining transaction

information that can contribute to analytical reporting. Historical records can help the business examine sales trends and identify products that are frequently purchased. This provides additional value beyond simply recording the completion of individual sales transactions.

Product demand is another area supported by the reviewed concepts. When product information is connected with transaction records, the business can examine how frequently different products appear in completed sales. This can provide information about frequently purchased products based on recorded transactions For Luna's Arrozcaldo, such information can contribute to understanding customer purchasing patterns and monitoring sales performance. The proposed analytical reporting component therefore gives stored sales data a practical purpose.

Customer purchasing behavior is also connected to the proposed system. When customer information is associated with appropriate transaction records, the system can provide a basis for examining purchasing activities. Repeated transactions may provide information that can help identify recurring purchasing behavior. This supports one of the analytical purposes of the proposed system, which is to analyze customer purchasing behavior using historical transaction data. The literature therefore supports the idea that organized transaction information can be used to provide a broader understanding of business activities.

The reviewed literature concerning business intelligence and analytical systems further supports this approach. Wixom and Watson (2010) discussed the role of business intelligence in helping organizations collect, manage, analyze, and present information for decision-making. This concept is closely related to the analytical reporting component of the proposed system. The system is not intended only to store transaction records; it is also intended to organize those records into information that can be reviewed by management. This gives the sales data a broader role in supporting business monitoring and decision-making. Analytical reporting also provides a connection between everyday operations and management needs. During a normal business day, personnel are primarily concerned with processing sales and recording transactions. Management, however, may need to look beyond individual transactions and examine overall sales performance. The proposed system connects these two needs by using recorded sales transactions as the basis for analytical reports. This allows operational information to contribute to management-level understanding of the business.

The reviewed literature also supports the importance of system usability. Davis (1989) explained that perceived usefulness and perceived ease of use influence users' acceptance of information technology. This is relevant to Luna's Arrozcaldo because personnel will interact with the proposed system during actual business operations. The system should therefore provide functions that are understandable and useful for their responsibilities. A system that is difficult to understand may affect how effectively users can perform their tasks, even if the underlying technology is functional.

The importance of usability is particularly relevant to a food-service business. Sales transactions may occur continuously during business operations, requiring personnel to enter information efficiently. The proposed system should therefore be designed around the actual Another common point among the reviewed materials is the importance of matching technology with actual organizational needs. Digital systems are more useful when their functions address real problems rather than simply introducing technology for its own sake. The proposed system responds to the need for organized sales processing, centralized sales information, and analytical reporting. Its features are therefore connected to the business processes identified for Luna's Arrozcaldo. This makes the proposed system relevant to the actual context of the study.

The reviewed studies also provide support for the use of web-based applications. Gaingan (2025) demonstrated a local example involving a web-based market goods trading system, while Encarnacion and Genodia (2016) demonstrated the application of an online ordering system in a local food-business context. These examples support the practicality of web-based approaches for managing business activities. The proposed system similarly uses a web-based environment to provide access to its sales-processing and management functions.

The combination of a web-based interface and centralized data storage can also support access to business information through a common system environment. Users can interact with the application while the system manages the underlying data. This arrangement supports the separation between the user interface, application processing, and data management described in the system architecture.

The reviewed literature therefore provides and frequently purchased products can be reviewed through organized reports. This does not automatically determine what decisions the business should make, but it provides information that can serve as a basis for decision-making. This distinction is important to the proposed system because its role is to provide organized information and analytical reports that can support management. The local context makes this especially relevant to Luna's Arrozcaldo. Philippine food businesses may have operational conditions that differ from larger organizations with extensive technological resources.

The reviewed local studies demonstrate that digital solutions can be applied within local business environments when they are designed around actual workflows and available resources. The proposed system follows this general direction by focusing on the specific sales-management and reporting requirements of Luna's Arrozcaldo.

The reviewed literature also reveals that digital transformation is not limited to replacing paper-based processes. A meaningful digital system can provide additional functions that were difficult to perform through manual records. For the proposed system, this includes using accumulated transaction data for analytical reporting. Thus, the system does not only digitize sales recording but also creates a structured basis for examining sales activities. This distinction strengthens the relevance of the proposed system to the identified business needs.

Another important point from the reviewed literature is that information should be presented in a form that users can understand. Analytical reporting becomes more useful when users can easily interpret the information provided. The proposed system therefore includes reports and dashboards intended to present relevant sales information in an accessible manner. This supports the broader principle of transforming stored data into information that can assist users in understanding business performance.

Across the reviewed sources, several connected principles emerge: useful information depends on reliable source records; software should be designed around user needs; data relationships should be controlled; security should be included throughout development; and quality should be evaluated against defined criteria. Together, these principles support the project's emphasis on sales processing, centralized management, and analytical reporting. They also show why a polished interface alone is not enough to establish system quality.

The literature informs the design of a traceable data path from a sale to a report. A transaction should have a stable identifier, its line items should preserve quantities and applicable prices, and branch and date information should be available for filtering. The reporting logic should use documented definitions so that the same query conditions yield the same interpretation. These requirements connect database design with the management purpose of the dashboard.

A gap addressed by the capstone is the need to apply these general concepts to the workflow of a specific local food business. The project brings together point-of-sale activity, branch-aware records, inventory visibility, customer information, and reporting in a single web application. The degree to which this integration is complete should be demonstrated in Chapter 4 through screenshots, tests, schema descriptions, and observed results rather than inferred from the intended architecture.

The synthesis also identifies limitations that remain relevant after development. Data quality depends on user behavior and validation; analytics depend on accurate and sufficiently complete records; and centralized access depends on hosting and connectivity. These limitations support practical recommendations for training, security review, data-quality checks, backup verification, and continued user feedback. They also define priorities for future research and development.

REFERENCES

[9] G. Vial, “Understanding digital transformation: A review and a research agenda,” The Journal of Strategic Information Systems, vol. 28, no. 2, pp. 118–144, 2019.

[10] P. C. Verhoef et al., “Digital transformation: A multidisciplinary reflection and research agenda,” Journal of Business Research, vol. 122, pp. 889–901, 2021.

[11] D. Buhalis and R. Leung, “Smart hospitality—Interconnectivity and interoperability towards an ecosystem,” International Journal of Hospitality Management, vol. 71, pp. 41–50, 2018.

[12] F. D. Davis, “Perceived usefulness, perceived ease of use, and user acceptance of information technology,” MIS Quarterly, vol. 13, no. 3, pp. 319–340, 1989.

[13] J. Highsmith and A. Cockburn, “Agile software development: The business of innovation,” Computer, vol. 34, no. 9, pp. 120–127, 2001.

[14] S. Negash, “Business intelligence,” Communications of the Association for Information Systems, vol. 13, no. 1, pp. 177–195, 2004.

PHINMA UNIVERSITY OF ILOILO
College Of Information Technology Education

CHAPTER III

METHODOLOGY

3.1 Research Design

The study will use a developmental research design, focusing on the design, development, testing, and evaluation of the proposed Digital Sales Processing System for Luna's Arrozcaldo. The research will examine the existing sales recording, transaction management, and reporting practices of the business to identify areas where a digital system can improve organization, accuracy, and efficiency. The findings from the existing process will serve as a basis for developing a centralized system that can record sales transactions, manage sales information, and provide analytical reports to support business decision-making.

The study uses a developmental research design because its main output is a working information system. Developmental research is appropriate when researchers identify a real problem, design a solution, develop a functional product, and evaluate whether the product meets the intended needs.

For this study, the development process begins with understanding Luna's Arrozcaldo's existing sales and management activities. The researchers identify problems in manual sales recording, branch monitoring, inventory information, customer records, and reporting. These findings are translated into system requirements and then into a digital solution.

The developmental process includes analysis, design, development, testing, deployment, and evaluation. The process is iterative, which means that feedback from users can be used to improve the system before the final version is delivered. This approach is consistent with the project documentation, which identifies developmental research and describes the system as being refined through evaluation and user feedback.

The study will use a developmental research design, focusing on the design, development, testing, and evaluation of the proposed Digital Sales Processing System with Centralized Sales

Management and Analytical Reporting for Luna's Arrozcaldo. This research design is appropriate because the study involves creating a working system that addresses identified needs in the business's sales-related processes. Rather than focusing only on describing an existing situation, the study will use the gathered information as a basis for developing a practical technological solution that can be evaluated based on its intended functions and usefulness.

The developmental research approach will begin by examining the existing sales-related processes of Luna's Arrozcaldo. The researchers will consider how sales information is handled, recorded, organized, and used for reporting and management purposes. This stage is important because understanding the existing process allows the researchers to identify areas that may benefit from a centralized digital system. The information gathered will help establish the requirements that the proposed system needs to address, particularly those related to sales processing, sales management, and analytical reporting.

The study will also focus on determining the information and processes that should be represented in the proposed system. Sales transactions contain information that can be organized and stored for later use in reporting and analysis. By providing a centralized approach to managing these records, the proposed system is intended to make sales information easier to organize and retrieve. The development will therefore consider how transaction information can move from the sales processing stage into centralized records that can later support analytical reports.

Another important part of the research design is the development of the system based on identified requirements. The researchers will translate the information gathered from the existing business process into system functions and an organized system structure. The proposed system will provide a web-based environment where authorized users can process and manage sales information. The system will use HTML, CSS, and JavaScript for the front-end interface and PHP for the back-end processing, allowing the different parts of the application to work together according to the identified requirements.

The developmental process will also consider the relationship between operational sales activities and analytical reporting. Sales processing is not limited to recording individual transactions because the accumulated information can also provide useful information for management. Through organized sales records, the system can support the preparation of analytical reports that help present sales information in a more understandable manner. This supports the main purpose of the capstone, which is to connect daily sales processing with centralized management and reporting within one digital system.

The study will further involve testing and evaluation of the developed system. Testing is necessary to determine whether the developed functions operate as intended and whether the different parts of the system work together properly. The researchers will examine the system's ability to process sales information, manage centralized records, and produce the intended

analytical outputs. Any identified problems or areas requiring improvement can then be considered during the development and refinement of the system.

Evaluation will provide a basis for determining whether the proposed system is appropriate for its intended purpose. The system will be assessed according to relevant aspects such as functionality, usability, accuracy, security, and efficiency. Functionality will determine whether the system performs its intended operations, while usability will consider whether users can understand and navigate the system without unnecessary difficulty. Accuracy will focus on the proper recording and presentation of sales information, while security will consider the protection of system records and user information. Efficiency will consider whether the system can make sales processing, information management, and reporting more organized.

The developmental research design will therefore provide a structured basis for moving from the identification of an existing business need toward the creation and evaluation of a digital solution. Each part of the study will contribute to the overall development of the proposed system. The examination of the current process will provide the basis for the requirements, the requirements will guide the system design, the design will guide development, and the developed system will undergo testing and evaluation.

In addition, the research design will help ensure that the proposed system remains connected to the actual operations of Luna's Arrozcaldo. The system will not be developed independently of the business context; instead, its intended functions will be based on the sales processing and management needs identified by the researchers. This approach is important because a digital system should not only function technically but should also be relevant to the users and processes for which it is developed.

The study will ultimately produce a working prototype that demonstrates how digital technology can be applied to improve the organization of sales-related information. The proposed system is intended to provide a centralized method for handling sales records while supporting the generation of analytical reports. Through this approach, the research will evaluate whether the developed application can provide a practical and organized solution for the sales management and reporting needs of Luna's Arrozcaldo.

Overall, the use of a developmental research design is suitable for this capstone because the primary output is a developed technological system rather than a purely theoretical result. The design allows the researchers to move systematically from understanding the existing process, identifying requirements, designing and developing the application, testing its functions, and evaluating its overall suitability. In this way, the research design directly supports the goal of developing a Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo.

3.2 System Development Methodology

This study adopted the **Agile Software Development Methodology** in designing and developing the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. Agile is an iterative and incremental software development methodology that emphasizes continuous improvement, regular evaluation, flexibility, and collaboration throughout the development process. Unlike traditional development approaches that generally complete each phase in a fixed sequence, Agile allows the system to be developed through repeated cycles, enabling the researchers to review completed functions, identify necessary changes, and improve the system based on the identified needs.

The Agile methodology was selected because it provides flexibility during the development of the proposed system. The operational requirements of Luna's Arrozcaldo may require adjustments as the researchers gain a better understanding of the business's sales processing and management activities. Through an iterative approach, the researchers can develop the system gradually, evaluate individual functions, and make necessary improvements before proceeding to the next development activities. This approach helps ensure that the developed system remains aligned with the actual sales and management needs of Luna's Arrozcaldo.

The Agile Software Development Methodology followed in this study consists of six major phases: Planning, Requirements Analysis, System Design, Development, Testing, and Deployment and Maintenance. Each phase contributes to the development of the proposed digital sales processing system and helps ensure that its functions support centralized sales management and analytical reporting.

3.2.1 Planning

The planning phase served as the foundation of the development of the proposed system. During this phase, the researchers identified the main purpose of the project, established its scope, determined the expected system functions, and identified the resources needed for development. The researchers focused on understanding how a digital system could improve the recording, management, and analysis of sales information at Luna's Arrozcaldo.

The researchers initially considered the existing sales-related activities of Luna's Arrozcaldo to determine the areas that could benefit from computerization. Attention was given to the recording of sales transactions, organization of sales information, monitoring of sales records, and preparation of reports. The researchers also considered how centralized management of sales information could help make business records more organized and easier to review.

Based on the identified needs, the researchers prepared the development plan and established the major functions to be included in the proposed system. The planning phase also considered possible challenges that could affect development, including changes in system requirements, data organization, development time, and technical limitations. Identifying these concerns during the planning stage allowed the researchers to prepare an organized approach for the succeeding phases of system development.

During the planning phase, the researchers defined the project goals, scope, main activities, timeline, and resources needed to develop the Digital Sales Processing System for Luna's Arrozcaldo. The team reviewed the existing sales-related activities to identify which tasks needed improvement and which system functions should be prioritized. Because the study adopted Agile, the development plan remained adjustable when feedback or newly identified needs required changes. This approach is consistent with the Principles behind the Agile Manifesto, which emphasize delivering useful software regularly and responding to changing requirements.

3.2.2 Requirements Analysis

The requirements analysis phase focused on identifying the specific needs of Luna's Arrozcaldo in relation to sales processing, sales management, and analytical reporting. The primary purpose of this phase was to understand the existing processes and determine the functional and non-functional requirements that the proposed digital sales processing system should satisfy.

The researchers gathered information regarding the business's sales activities and considered the processes involved in recording and managing transactions. The analysis focused on how sales information is recorded, how transaction records are organized, how sales performance can be monitored, and how information can be presented through reports and analytical views. The researchers also considered the information needed by the users to perform sales-related activities efficiently.

The information gathered during this phase was organized into system requirements. The functional requirements included digital sales transaction processing, centralized sales management, recording and storage of sales information, sales monitoring, analytical reporting, dashboard presentation, and analysis of customer purchasing behavior. The system also considers the monitoring of product demand and sales trends based on available transaction data.

The non-functional requirements focused on important characteristics of the system, including usability, reliability, security, accuracy, performance, and efficiency. These requirements were considered to ensure that the system would not only perform its intended functions but would also provide users with an understandable and reliable platform for managing sales information.

The requirements identified during this phase served as the primary basis for designing the system structure, database organization, interface, and major system functions.

The requirements analysis phase identified and organized the sales-processing needs that the proposed system had to support. Functional requirements included digital transaction recording, centralized sales information, dashboards, and analytical reports, while non-functional requirements addressed usability, reliability, security, accuracy, and performance. The researchers reviewed these requirements for clarity, feasibility, and whether each could be evaluated during system testing. This process aligns with NASA's Software Requirements Analysis guidance, which explains why requirements should be clear, feasible, and verifiable before guiding further development.

3.2.3 System Design

Following the requirements analysis, the researchers proceeded to the system design phase. This phase focused on transforming the identified requirements into a structured design that would serve as a guide for developing the Digital Sales Processing System for Luna's Arrozcaldo.

During this phase, the researchers designed the overall system structure, database organization, user interface, navigation flow, and relationships between the different system processes. System diagrams and flowcharts were prepared to illustrate how users interact with the system, how sales transactions are processed, how information is stored, and how recorded data is used for reporting and analysis.

The user interface was designed with simplicity and ease of use as important considerations. The researchers organized the system's navigation, transaction screens, sales records, dashboards, and analytical reports in a manner that allows users to access information and perform sales-related tasks more easily. Clear labels, readable information, and organized screens were considered to help users understand the system without requiring complicated procedures.

The database design focused on properly organizing the information required by the system. Data related to users, products, sales transactions, sales details, and analytical information were structured to support accurate recording and retrieval of information. The relationships among the stored data were also considered so that sales transactions could be properly connected with the products involved and could later be used for generating reports and identifying purchasing and sales patterns.

The system design therefore provided the framework that guided the researchers in developing a centralized system capable of connecting sales processing, sales management, and analytical reporting.

During the system design phase, the identified requirements were organized into a practical structure for the website. The researchers planned the user interface, navigation, database tables, and relationships connecting users, products, sales transactions, and sales details. These design decisions aimed to keep records organized and enable transaction data to be retrieved for

centralized management and analytical reporting. This process is consistent with NASA's Systems Engineering Handbook, which discusses system architecture, design, and verification planning as parts of developing and organizing a system.

3.2.4 System Development

During the development phase, the researchers translated the completed system design into a functional web-based application. Development was performed incrementally following the principles of Agile development, wherein system functions were developed, reviewed, and improved throughout the development process.

The system was developed using PHP as the primary programming language for the application, with HTML, CSS, and JavaScript used for the front-end interface and interactive functions. The development environment was used to create and organize the different components of the proposed digital sales processing system.

The researchers developed the system according to its major functions. The sales processing function was developed to allow sales transactions to be recorded digitally and organized systematically. The centralized sales management function was developed to provide a more organized way of viewing and managing recorded sales information. The system also includes analytical reporting and dashboard functions that present sales information in a more understandable form.

The development also focused on using stored transaction data to support the analysis of customer purchasing behavior, frequently purchased products, product demand, and sales trends. Through these functions, the system can transform recorded sales transactions into useful information that can assist Luna's Arrozcaldo in reviewing its business performance and making informed decisions.

Throughout the development process, the researchers reviewed the completed functions and made necessary adjustments when problems or improvements were identified. This iterative process allowed the system to gradually become more aligned with the identified requirements of Luna's Arrozcaldo.

During system development, the researchers implemented the approved design as the web-based sales processing system. The developed functions covered recording sales transactions, organizing product and sales records, and presenting stored information through dashboards and analytical reports. The team reviewed the functions incrementally and made corrections when problems or opportunities for improvement were identified. The PHP Manual provides guidance for building dynamic web pages and forms, while the NIST Secure Software Development Framework recommends incorporating secure development practices throughout the software development process.

3.2.5 Testing

System testing was conducted throughout the development process to identify errors and verify whether the developed functions operated according to the identified requirements. Instead of waiting until the entire system was completed, individual functions were reviewed and tested during development so that problems could be identified and corrected earlier.

Functional testing was performed to determine whether the major functions of the system operated correctly. The researchers tested the recording of sales transactions, storage and retrieval of sales information, centralized sales management, product-related records, dashboard information, and analytical reports. The researchers also checked whether the information displayed by the system corresponded with the data recorded from sales transactions.

Integration testing was also conducted to verify whether the different parts of the system worked properly together. Particular attention was given to the relationship between sales transaction processing, stored sales records, and analytical reporting. The researchers verified whether recorded transaction information could be properly retrieved and reflected in the appropriate reports and analytical displays.

The system was also subjected to User Acceptance Testing (UAT) to determine whether the proposed system was understandable and suitable for its intended users. The users evaluated the system by performing relevant sales management tasks and reviewing the information produced by the system. Any issues identified during testing were considered for correction and improvement before the system was finalized.

Testing was conducted to check whether the system’s major functions operated according to the identified requirements. The researchers checked sales recording, retrieval of saved records, centralized sales views, dashboard information, and analytical reports for correct and consistent output. Integration checks focused on whether stored transaction data appeared correctly in the related management screens and reports. The OWASP Web Security Testing Guide also provides guidance for evaluating web applications and identifying potential security weaknesses.

3.2.6 Deployment and Maintenance

Following the completion of testing and the necessary improvements, the developed Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo was prepared for deployment. The deployment phase involved preparing the application and database environment, checking system accessibility, and ensuring that the developed functions were properly working before actual use.

The researchers also provided guidance regarding the basic operation of the system. This included the use of the sales processing functions, management of sales records, viewing of centralized

sales information, and interpretation of dashboards and analytical reports. Proper orientation was considered important so that users could understand how the system supports their regular sales-related activities.

Maintenance was considered necessary after deployment to help maintain the reliability and effectiveness of the system. Errors or minor issues discovered during use could be reviewed and corrected through system updates. Improvements could also be considered when changes in operational requirements or user needs were identified.

After testing and necessary corrections, the team deployed the website through Railway, a platform that provides application build and deployment workflows. The deployed website was made accessible through the project's Railway URL so intended users could access the system online. For database administration, the team used the AdminPHP interface to access and maintain stored sales and product information, while the phpMyAdmin documentation describes similar database administration functions. For ongoing maintenance, the team paid a total of $2 per month, divided equally among five members, with each member contributing $0.40 monthly.

The Agile methodology supports this continuing improvement because the system can be reviewed and refined through additional development cycles. This allows the proposed system to remain useful for Luna's Arrozcaldo while continuing to support digital sales processing, centralized sales management, and analytical reporting.

Overall, the six Agile phases provided an organized process for developing the proposed system. The methodology allowed the researchers to move from identifying the business requirements to designing, developing, testing, and preparing the system for use. Through this process, the study aimed to produce a system that provides a more organized approach to sales transaction recording, centralized sales management, and the generation of useful analytical information for Luna's Arrozcaldo.

Figure 3.1 Agile Software Development Methodology

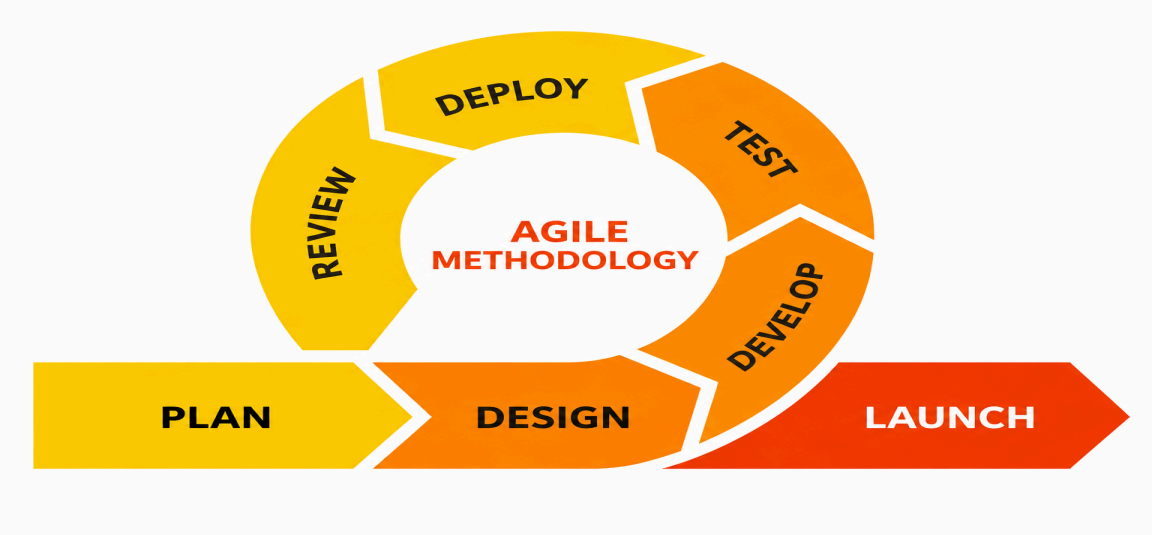


Figure 3.1 illustrates the Agile Software Development Methodology adopted in the study. Unlike traditional linear development models, Agile promotes continuous feedback and iterative improvement. Each completed module was evaluated by the client, allowing the researchers to implement revisions and enhancements throughout the software development life cycle.

3.3 System Architecture

The system follows a client-server structure with three practical layers: the user interface, the application or server layer, and the centralized database. This structure allows users to interact with the system through a browser while the backend handles business rules and database operations.

The developed system will follow a client-server architecture designed to support the main functions of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. The architecture provides a clear connection between the user interface, server-side processing, and centralized storage of sales information. Through this structure, users can interact with the system using a web browser, while the server processes the requests and manages the information needed by the application. The architecture is designed around the actual purpose of the capstone, which is to provide a digital approach to sales processing, centralized sales management, and analytical reporting.

The architecture is divided into three main parts: the client or front-end layer, the server or back-end layer, and the database layer. These components work together to allow information to move from the user's interaction with the system to the processing and storage of the information. The front end serves as the part of the system that users see and interact with, while the back end handles the processes that occur behind the interface. The database serves as the central location where important system information, particularly sales transaction records, is stored and retrieved when needed.

The use of a client-server architecture is appropriate for the proposed system because sales information needs to be processed and managed in an organized manner. Instead of depending only on separate manual records, the system provides a centralized environment where sales transactions can be recorded and accessed through the application. This allows the different sales-related processes to work together and provides a basis for generating useful information from recorded transactions.

Front-End Architecture

The front end represents the user-facing portion of the system. It will be developed using HTML, CSS, and JavaScript, which work together to create the structure, appearance, and interactive behavior of the web-based application. The front end provides the screens and controls that authorized users will use when performing sales-related activities.

HTML will be used to structure the different pages and elements of the system. It provides the basic structure for forms, buttons, tables, navigation areas, transaction information, and other elements that users need when interacting with the application. Through an organized HTML structure, the system can present sales-related information in a clear and understandable manner.

CSS will be used to organize the appearance and layout of the system. It helps maintain consistency in the arrangement of menus, forms, tables, buttons, dashboards, and other interface components. A properly organized interface is important because the intended users need to access sales functions and information without unnecessary difficulty.

JavaScript will provide interactive behavior on the front end. It can be used to support actions that require immediate interaction within the webpage, such as validating information entered into forms, responding to button actions, updating displayed information, and improving the overall interaction between the user and the system. These functions help make the web application more responsive and easier to use.

The front-end component is important because it serves as the connection between the user and the system's internal processes. When an authorized user enters information, such as a sales transaction, the information begins at the user interface. The information is then sent to the back end for processing before it is stored in the database. When the user requests information, such as

previous sales records or analytical reports, the front end displays the processed information returned by the back end.

Back-End Architecture

The back end serves as the processing component of the system and will be developed using PHP. It is responsible for handling requests received from the front end, processing sales-related information, communicating with the database, and returning the required results to the user interface.

PHP provides the server-side logic needed for the system to perform its major operations. When a user submits a sales transaction through the front end, the request is received and processed by the PHP back end. The back end checks and processes the submitted information before communicating with the database. After the information has been successfully processed, the appropriate response can be returned to the front end.

The back end also helps separate the visible interface from the internal processing of the system. Users do not need to directly interact with the database because the PHP application serves as the intermediary. This structure helps organize the system because the front end focuses on user interaction while the back end focuses on processing and managing the application's operations.

For the proposed system, the back end will support the processing of sales transactions and the management of centralized sales information. It will also handle requests related to retrieving sales records and preparing information needed by the analytical reporting functions. This allows transaction data recorded through the system to become useful information that can be viewed through reports and dashboards.

The server-side component also provides a central point for applying system rules and processing procedures. For example, when a sales transaction is submitted, the back end can process the transaction information according to the system's defined requirements before it is stored. This helps maintain consistency in the way sales information is handled throughout the application.

Database Layer

The database layer serves as the central storage component of the architecture. It stores the information processed by the application so that the records can be retrieved and used whenever necessary. For the proposed Digital Sales Processing System, the database is particularly important because the system depends on stored sales transaction data for centralized sales management and analytical reporting.

Sales information entered through the front end is processed by the PHP back end before being stored in the database. The stored information can later be retrieved when authorized users need to review previous transactions, monitor sales information, or generate analytical reports. This creates a continuous connection between sales processing and reporting.

The database also helps organize information so that the system can distinguish different types of records and maintain relationships between related information. Proper organization of the stored data is important because the system needs to retrieve accurate information when producing reports. If sales information is properly recorded and organized, the system can use historical transactions as a basis for identifying sales trends, frequently purchased products, and other useful patterns supported by the available data.

The centralized database therefore supports one of the major purposes of the capstone. Instead of treating every transaction as an isolated record, the system stores transaction information in a common location that can be accessed by the appropriate system functions. This makes it possible for sales processing, sales management, and analytical reporting to work together.

Interaction Between the Front End and Back End

The interaction between the front end and back end is an important part of the system architecture. When an authorized user performs an action on the web interface, the front end receives the user's input and sends the appropriate request to the PHP back end. The back end then processes the request and performs the necessary operation.

For example, during sales processing, the user enters the required transaction information through the front-end interface. The system sends the submitted information to the PHP back end. PHP processes the transaction and communicates with the database to store the information. Once the operation is completed, the result is returned to the front end so that the user can see the appropriate confirmation or updated information.

The same basic process applies when users need to retrieve existing information. A user may request a sales record or report through the front end. The request is sent to the PHP back end, which communicates with the database to retrieve the required information. The retrieved data is then processed as necessary and presented through the front-end interface.

This communication allows the system to operate as a connected application rather than as separate functions. The front end provides the interaction, the back end performs the processing, and the database provides the stored information needed by the system.

Sales Transaction Processing Flow

The architecture supports the digital processing of sales transactions through a series of connected steps. First, an authorized user accesses the sales processing interface through the web application. The user enters the necessary transaction information using the available fields and controls.

After the transaction information is submitted, the front end sends the information to the PHP back end. PHP receives the request and processes the information based on the requirements of the system. The back end then communicates with the database to record the transaction.

Once the transaction has been successfully stored, the system can retrieve the information for other functions. The recorded transaction becomes part of the centralized sales records. These records can later be used for reviewing sales activities and producing analytical information.

This process reduces the dependence on manually organizing individual sales records because the information is digitally recorded and stored through the system. It also creates a consistent source of information that can support the other functions of the proposed system.

Centralized Sales Management

Centralized sales management is another important function supported by the architecture. The system provides a common environment where authorized users can access and manage recorded sales information. Since the sales data is stored centrally, the system can retrieve information without requiring users to search through separate records.

The front end provides the interface for viewing and managing sales information, while PHP handles the requests and communicates with the database. This arrangement allows users to interact with sales records without directly accessing the underlying database.

Centralization also supports better organization of information. Sales transactions can be stored in a structured manner and retrieved when needed. This allows users to review previous transactions and examine sales information through the system's available management and reporting functions.

The centralized approach is particularly relevant to Luna's Arrozcaldo because the proposed system is intended to improve the way sales information is recorded and managed. The architecture provides a foundation where transaction processing and sales management are connected rather than treated as separate activities.

Analytical Reporting Component

The architecture also supports the analytical reporting component of the proposed system. The analytical reporting function depends on the transaction information stored in the database. Once sales transactions have been properly recorded, the system can retrieve and process the available data to produce useful reports and analytical displays.

The PHP back end serves as an important connection between stored transaction data and the reporting interface. When a user requests an analytical report, the system retrieves the necessary information from the database. The back end processes the requested information and provides the results to the front end, where they can be displayed in an understandable format.

The analytical information can support the examination of sales trends, product demand, frequently purchased products, and customer purchasing behavior, based on the transaction data available within the system. This allows the proposed application to move beyond simple transaction recording. The stored data can also become a source of information that helps users understand sales performance.

The reporting component therefore demonstrates how the architecture supports the complete purpose of the capstone. Sales information begins as a transaction entered through the front end, is processed by PHP, stored in the database, and later retrieved for analytical reporting. The three layers work together to transform individual transactions into organized business information.

Data Flow Within the Architecture

The overall data flow begins with the user and moves through the front end, back end, and database. The user interacts with the web application using the interface developed with HTML, CSS, and JavaScript. When an action requires processing or storage, the request is sent to the PHP back end.

PHP receives the request and determines the required operation. For transactions, it processes the submitted sales information and communicates with the database. For information requests, it retrieves the required records from the database and processes them for presentation.

The database provides the stored information required by the back end. After completing the necessary operation, the back end sends the result back to the front end. The front end then displays the information to the user.

The flow can therefore be summarized as:

User → Front End → PHP Back End → Database → PHP Back End → Front End → User

For a sales transaction, the flow represents the movement of information from transaction entry to processing and centralized storage. For analytical reporting, the flow works in the opposite direction as stored information is retrieved, processed, and presented to the user.

Security and Controlled Access

The client-server structure also provides a basis for controlling access to system functions. Since users interact with the system through the web interface and the server handles the actual processing, access to the underlying database can be managed through the application.

Authorized users can be provided access to the functions necessary for their responsibilities. The back end serves as the processing point for requests and can apply the appropriate system rules before allowing an operation to proceed. This helps protect the integrity of the sales information managed by the application.

Security is particularly important because the system stores business-related information. Proper handling of user access and transaction information helps reduce the possibility of unauthorized changes to sales records. The architecture therefore considers security as part of the overall operation of the system rather than treating it as a separate function.

Efficiency and System Organization

The proposed architecture also supports efficiency by assigning different responsibilities to the front end, back end, and database. The front end handles user interaction, PHP handles application processing, and the database handles information storage and retrieval. Separating these responsibilities makes the system easier to organize and maintain.

For Luna's Arrozcaldo, this structure can help make sales-related activities more systematic. Instead of manually handling sales records and preparing information separately, the system connects the recording and management of sales information with analytical reporting.

The architecture also makes it easier to identify where a problem may occur. For example, an issue with the appearance or interaction of a page can be addressed within the front-end components, while an issue involving transaction processing can be examined within the PHP back end. Problems involving stored information can be investigated within the database component. This separation provides a more organized foundation for maintaining the application.

Overall System Architecture

Overall, the client-server architecture provides the technical structure needed to connect the major functions of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo**. The HTML, CSS, and JavaScript front end provides the interface through which authorized users interact with the system. The PHP back end processes requests, handles sales-related operations, and communicates with the database. The database centrally stores the information required for sales management and reporting.

The architecture creates a continuous flow of information from transaction entry to centralized storage and analytical reporting. Through this connection, recorded sales transactions can be organized and later used to produce useful information about sales performance and purchasing patterns. This supports the main objective of developing a digital system that can improve the way **Luna's Arrozcaldo** handles sales information.

In summary, the architecture is designed to ensure that the different parts of the proposed system work together as one application. The front end allows users to interact with the system, the PHP back end manages the processing, and the database provides centralized storage. Their combined operation supports digital sales processing, centralized sales management, and analytical reporting, which are the core functions reflected in the capstone title.

3.4 Tools and Technologies Used

The following tools and technologies were used in developing the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo**. Each technology was selected based on its role in building the food ordering, sales processing, centralized data management, and reporting functions of the system.

Frontend: **HTML**

- HTML was used to create the basic structure and content of the system's web pages.
- It was used to organize important sections such as the food menu, ordering page, sales transaction forms, product information, and reports.
- HTML provided the different fields and buttons needed for users to enter and process food orders.

- It also helped organize information displayed to authorized users, such as product names, prices, quantities, and transaction details.

- The structured HTML pages served as the foundation that allowed CSS and JavaScript to add the system's appearance and interactive functions.

Frontend: CSS

- **CSS** was used to design and arrange the visual appearance of the food ordering system.

- It was applied to menus, buttons, forms, tables, navigation areas, and dashboard components to make the interface easier to understand.

- **CSS** helped maintain a consistent layout throughout the different pages of the system.

- It was also used to improve the readability of sales records, product information, and analytical reports.

- The use of CSS allowed the researchers to create an interface that is simple and practical for users who will perform regular food ordering and sales-related activities.

Frontend: JavaScript

- **JavaScript** was used to add interactive functions to the web-based system.

- It supported actions such as selecting food products, changing quantities, calculating order totals, and responding to user inputs.

- **JavaScript** helped make the ordering process more responsive by allowing certain information to be updated without requiring users to repeatedly perform manual calculations.

- It was also used to support interactive elements of the sales interface and dashboard.

- Through these functions, **JavaScript** helped connect the user's actions on the frontend with the processes required for food ordering and sales management.

● Backend: PHP

- PHP was used as the main backend programming language of the system.

- It handled the processing of information submitted through the food ordering and sales interfaces.
- PHP was responsible for processing sales transactions, receiving order information, and communicating with the database.
- It helped manage the flow of information between the user's web interface and the centralized database.
- PHP also supported the system's sales management functions by processing stored transaction information that could later be used for reports and analysis.
- The backend provided the necessary processing layer so that sales information was not only displayed on the website but could also be properly handled and stored.

Database: PostgreSQL (via Supabase)

- PostgreSQL, accessed through Supabase, was used as the database for storing the system's organized records.
- It was used to store important information related to food products, prices, customer orders, sales transactions, and other necessary sales records.
- The database provided a centralized location for transaction information, allowing the system to retrieve records when needed.
- Relationships between related records helped organize sales information and reduce unnecessary duplication of data.
- Stored transaction data could be used as the basis for sales monitoring, analytical reporting, and identifying purchasing or sales patterns.
- Supabase provided the platform through which the PostgreSQL database could be managed and connected with the developed system.
- This database setup supported the main purpose of the capstone by keeping sales-related information organized and available for the system's management and reporting functions.

Development Tool: Visual Studio Code

- Visual Studio Code was used as the main development environment for creating and organizing the system.
- It was used to write and edit the HTML, CSS, JavaScript, and PHP source code.
- The researchers used it to organize the different files and components of the food ordering system during development.
- It also provided a convenient environment for checking and revising the code as different system functions were developed.
- Using Visual Studio Code allowed the researchers to work on the frontend and backend components in an organized manner.
- It supported the continuous development and improvement of the system as the researchers tested and refined the different functions.

Overall Use of the Technologies

These technologies worked together to form the system's main structure. HTML, CSS, and JavaScript handled the user-facing food ordering interface, while PHP processed the requests and sales transactions on the backend. PostgreSQL through Supabase provided centralized storage for the system's sales and product information, while Visual Studio Code served as the development environment for creating and managing the system. Together, these tools supported the capstone's focus on digital sales processing, centralized sales management, and analytical reporting for Luna's Arrozcaldo.

3.5 Data Gathering Techniques

The study gathers information mainly through interviews, observation of existing business processes, review of current records and procedures, system walkthroughs, and feedback from representative users. These techniques help the researchers understand what users actually do, what information they need, and what problems the digital system should solve.

Interviews

Interviews are used to obtain direct explanations from the business owner, branch managers, and staff about sales processing, inventory monitoring, customer records, promotions, reporting, and access needs.

In this study, interviews were used to gather additional information from Luna's Arrozcaldo's business owner Ma'am Helen, branch managers, and staff regarding their sales operations and reporting needs. The researchers focused on the difficulties encountered in recording transactions, organizing sales information, and monitoring business performance. Follow-up questions helped clarify the participants' answers and identify requirements that needed to be considered in developing the proposed system. This approach follows the guidance on conducting structured, in-depth interviews to understand users' experiences and needs provided by the UK Government's User Research Guide

Observation

Observation is used to understand the actual flow of work at the point of sale. The researchers pay attention to how transactions are recorded, how totals are checked, how records are summarized, and how information is transferred for management review.

Observation allowed the researchers to examine how sales-related activities were performed during regular business operations at Luna's Arrozcaldo. The researchers paid attention to transaction recording, checking sales totals, organizing records, and preparing information for management review. These observations helped identify actual work procedures and possible difficulties that might not be fully explained during interviews. The method is supported by the UK Government's guidance on contextual research and observation, which explains how observing people in their usual working environment can reveal their processes and challenges

Document and Record Review

Existing sales records and related procedures are reviewed to identify the information fields that must be represented in the digital system. This helps ensure that the database and reports reflect the business's real information needs.

Document and record review helped the researchers examine existing sales records and procedures relevant to the proposed Digital Sales Processing System. The review focused on the information recorded in transactions, the organization of sales documents, and the details needed for centralized sales management and analytical reporting. The researchers used these materials to identify essential data fields and determine how records should be organized in the proposed database. This technique is consistent with the Centers for Disease Control and Prevention's

guidance on gathering credible evidence, which identifies document and record reviews as a way to obtain background information and answer specific research questions.

User Feedback

Feedback from representative users is gathered during prototype review and testing. This is important because users can identify practical issues that may not be visible from documentation alone. The source project records user acceptance testing with representative administrator, branch manager, and cashier or counter staff users.

User feedback was gathered during the review and testing of the proposed system to understand whether its functions met the needs of the intended users. The feedback focused on the usability of the interface, the clarity of sales information, the accuracy of displayed records, and the usefulness of the dashboards and analytical reports. The researchers considered the comments and suggestions when identifying areas that required correction or improvement. This approach is consistent with the UK Government's guidance on user research for live services, which recommends collecting user feedback and evaluating people's experiences to identify problems and improve services.

3.6 Testing Procedures

Testing is performed to verify that the system functions correctly, that related modules work together, and that users can perform their assigned tasks. The following testing activities are used:

3.6.1 Unit Testing

Unit Testing will be conducted to examine the individual functions and components of the system separately. This procedure will allow the researchers to check whether each developed feature performs its intended task before the different components are combined into the complete system. The testing will focus on the frontend functions developed using HTML, CSS, and JavaScript, as well as the backend processes developed using PHP.

For the food ordering function, individual components will be checked to determine whether food products can be properly selected, quantities can be entered, and order information can be processed correctly. The researchers will also check whether the system calculates the appropriate order amount based on the selected products and quantities. Input fields, buttons, menus, and other interactive components will be tested to identify incorrect responses or unexpected behavior.

The PHP backend functions will also be tested individually. The researchers will verify whether information submitted through the frontend is properly received and processed by the backend. Functions responsible for handling sales transactions will be checked to ensure that the information follows the expected processing flow. Database-related functions will also be examined to determine whether transaction information is properly stored and retrieved from the PostgreSQL database through Supabase.

Unit Testing will also help identify errors at an early stage of development. When an individual function produces an incorrect result, the researchers can locate the specific component responsible for the problem and make the necessary revisions. This approach helps prevent small errors from affecting other parts of the system. After corrections are made, the affected function will be tested again to determine whether the problem has been properly addressed.

The testing will cover important individual functions such as product selection, order entry, quantity adjustment, transaction processing, data storage, data retrieval, and report-related functions. The researchers will record identified errors and check the results after corrections. Through this procedure, each component can be examined before being considered ready for integration into the complete system.

3.6.2 System Testing

System Testing will be conducted after the individual components have been tested and integrated. Unlike Unit Testing, which examines individual functions, System Testing will evaluate the developed application as a complete food ordering and sales processing system. The purpose is to determine whether the different parts of the system work together properly and support the main objectives of the capstone.

The researchers will test the complete flow of a sales transaction, beginning with the selection of food products and continuing through order processing and recording. The transaction information will be checked to determine whether it is properly handled by the PHP backend and stored in the PostgreSQL database through Supabase. The researchers will also verify whether previously recorded information can be retrieved correctly when needed for sales management and reporting.

The connection between the frontend, backend, and database will receive particular attention during System Testing. The HTML, CSS, and JavaScript interface must properly communicate the user's actions, while PHP must correctly process the submitted information and communicate with the database. The database must then store the appropriate information so that the system can use the recorded data for centralized sales management and analytical reporting.

The researchers will also test the reporting functions using recorded sales information. The purpose is to determine whether the system can properly use stored transaction data to present

useful sales information. The displayed information will be checked against the corresponding transaction records to ensure consistency. This is important because inaccurate stored or displayed information could affect the usefulness of the system's analytical reports.

System Testing will also consider the overall usability and flow of the application. The researchers will check whether users can move between the necessary pages, enter information correctly, process food orders, and access relevant sales information without encountering unnecessary problems. Errors encountered during testing will be documented, reviewed, and corrected before proceeding to User Acceptance Testing.

Integration between the major functions will also be examined. For example, information entered during food ordering should follow the expected transaction process and become available as a sales record. The researchers will check whether the information remains consistent as it moves from the frontend to the PHP backend and eventually to the centralized database. This ensures that the system operates as one connected application rather than as separate functions.

3.6.3 User Acceptance Testing

User Acceptance Testing (UAT) will be conducted to determine whether the completed system is acceptable and suitable for its intended use at Luna’s Arrozcaldo. This testing will involve the intended users performing relevant activities using the developed system. The purpose is to obtain practical feedback regarding the system's functionality, ease of use, and ability to support food ordering, sales processing, centralized sales management, and analytical reporting.

During UAT, users will be asked to perform common activities that represent the actual use of the system. These activities may include selecting food products, entering order information, processing sales transactions, checking recorded information, and viewing available sales reports. The users' experience while performing these activities will help determine whether the system is understandable and practical for its intended purpose.

The researchers will observe whether users can complete the required tasks using the provided interface. Particular attention will be given to the clarity of the ordering process, organization of information, ease of navigation, and accessibility of sales-related records. Feedback will also be gathered regarding any difficulties users experience while using the system.

The results of User Acceptance Testing will provide a basis for identifying areas that may still require improvement. If users identify problems in the ordering process, transaction recording, information presentation, or reporting functions, the researchers will review the identified concerns and make appropriate revisions within the scope of the developed system. After necessary adjustments, the affected functions may be tested again to confirm that the improvements work as intended.

Overall, the three testing procedures will provide different levels of evaluation for the developed system. Unit Testing will verify individual functions, System Testing will examine how the complete system and its components work together, and User Acceptance Testing will determine whether the system is practical and acceptable for its intended users. These procedures will help ensure that the developed system supports the capstone's primary purpose of providing digital sales processing, centralized sales management, and analytical reporting for Luna’s Arrozcaldo.

Testing is repeated after important corrections so that identified errors are not simply recorded but are actually addressed. This iterative approach is consistent with Agile development and the project's documented use of user feedback.

3.7 Evaluation Criteria

The system is evaluated using practical criteria that match its purpose as a sales and management information system.

CRITERIA	DESCRIPTION
Functionality	The system performs the required sales, management, database, and reporting functions correctly.
Usability	Users can understand the interface, navigate the system, and complete common tasks with reasonable effort.
Efficiency	The system supports faster and more organized processing than the manual approach.
Security	Access to system functions and sensitive information is controlled according to user roles.
Accuracy	Sales totals, transaction details, and generated reports reflect the recorded data correctly.

SUPPORTING DIAGRAMS

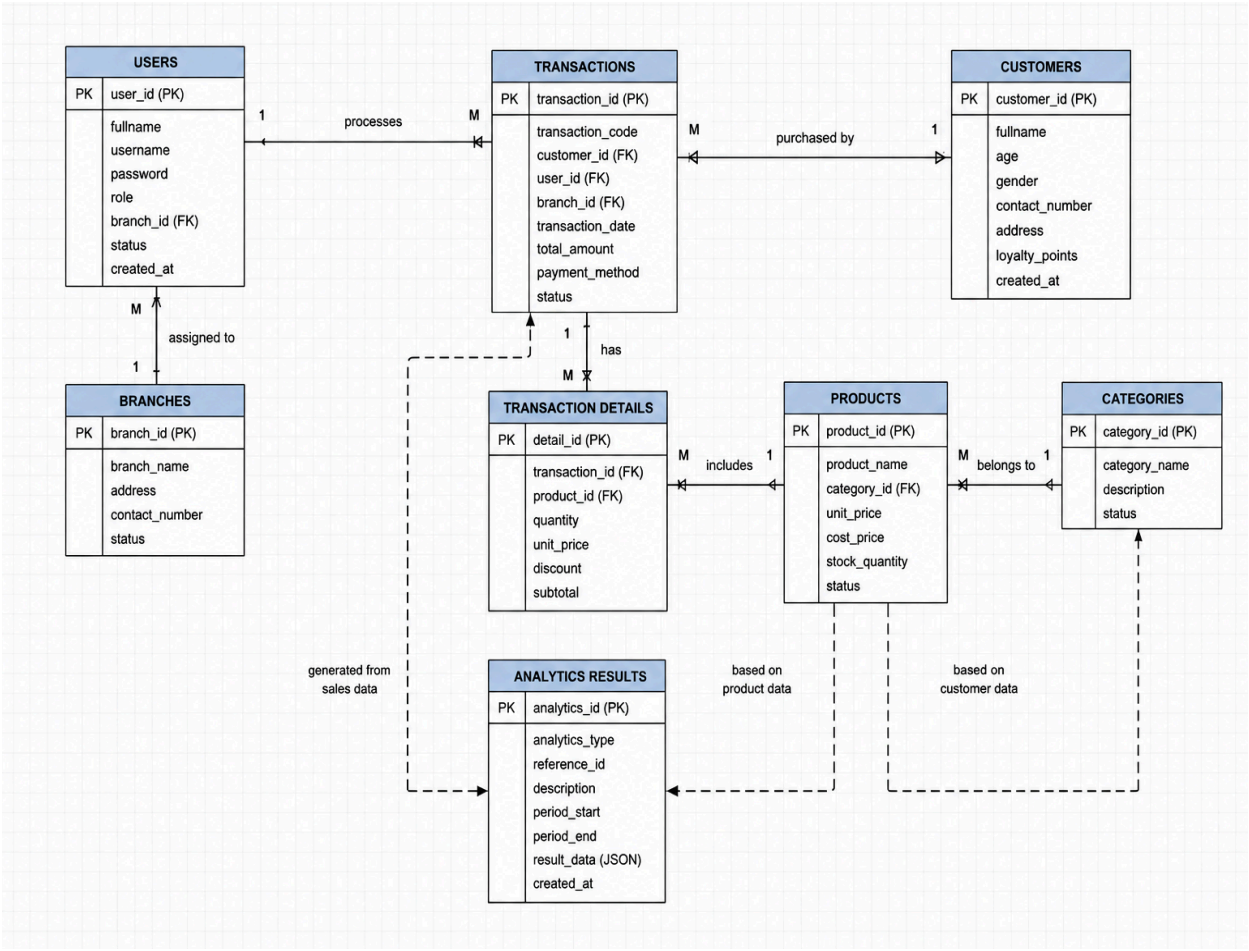
The following simplified diagrams follow the organization of the provided capstone proposal guide and are based on the actual system structure and corrected ERD for Luna's Arrozcaldo.

Data Flow Diagram (DFD – Level 0)

ACTOR	FLOW	PROCESS	FLOW	DATA STORE
Business Owner / Manager	views	Digital Sales System	stores	Centralized Database
Cashier	records	POS / Sales Processing	updates	Sales / Transaction Data

The **Level 0 Data Flow Diagram** illustrates how information moves between the authorized user, the Digital Sales Processing System, and the centralized PostgreSQL database through Supabase. The user submits food ordering and sales information through the system, where PHP processes the submitted data before it is stored in the database. The system can retrieve the stored transaction information for centralized sales management and analytical reporting, allowing recorded sales data to be transformed into organized information that can be presented to authorized users.

Entity Relationship Diagram – Main Entities



ENTITY	KEY INFORMATION	MAIN RELATIONSHIP
USERS	user_id, fullname, username, password, role, branch_id	A branch has many users; a user processes many transactions.
BRANCHES	branch_id, branch_name, address, contact_number, status	A branch has many users and many transactions.
CUSTOMERS	customer_id, fullname, age, gender, contact_number, address, loyalty_points	A customer can have many transactions.

ENTITY	KEY INFORMATION	MAIN RELATIONSHIP
TRANSACTIONS	transaction_id, transaction_code, customer_id, user_id, branch_id, date, total_amount, payment_method, status	A transaction belongs to one customer, user, and branch and has many details.
TRANSACTION DETAILS	detail_id, transaction_id, product_id, quantity, unit_price, discount, subtotal	Each detail belongs to one transaction and one product.
PRODUCTS	product_id, product_name, category_id, unit_price, cost_price, stock_quantity, status	A product belongs to one category and can appear in many transaction details.
CATEGORIES	category_id, category_name, description, status	A category can contain many products.
ANALYTICS RESULTS	analytics_id, analytics_type, reference_id, description, period_start, period_end, result_data, created_at	Stores generated analytical results based on recorded business data.

The ERD shows the main relational structure of the system. Transactions connect users, branches, and customers, while transaction details connect each sale to the products purchased. Products are organized under categories, and analytical results are generated from the business data used for reporting.

Use Case Diagram

21. Administrator – manages users, branches, products, categories, customers, promotions, and network-wide reports.

22. Branch Manager – monitors branch sales, inventory-related information, staff activity, and branch reports.

23. Cashier – logs in, processes sales, records transaction details, and performs the functions needed at the POS.

24. Management Reporting – retrieves stored sales information and presents summaries, comparisons, and trends.

25. Centralized Data Management – keeps authorized users connected to the same source of sales and operational information.

System Processing Flow

1. The user logs in using an authorized account.

2. The system identifies the user's role and available permissions.

3. The cashier records the customer's purchase through the POS.

4. The system validates the transaction and calculates the required totals.

5. The transaction and its details are stored in the centralized PostgreSQL database.

6. Related operational information can be updated according to the transaction and system rules.

7. Authorized management users can view centralized sales information.

8. The analytical reporting component organizes stored data into reports, charts, and summaries.

9. Users use the generated information to support daily monitoring and business decisions.

Chapter 3 Summary

This chapter described the developmental research design and Agile methodology used to create the system. It explained the three-layer system architecture, the actual technologies used, the data gathering methods, the testing procedures, and the evaluation criteria. It also presented simplified supporting diagrams based on the system structure. The methodology provides a clear path from the identified manual sales problems to a working digital system that supports centralized sales management and analytical reporting.

REFERENCES

[15] J. Highsmith and A. Cockburn, “Agile software development: The business of innovation,” Computer, vol. 34, no. 9, pp. 120–127, 2001.

[16] F. D. Davis, “Perceived usefulness, perceived ease of use, and user acceptance of information technology,” MIS Quarterly, vol. 13, no. 3, pp. 319–340, 1989.

[17] S. Chaudhuri, U. Dayal, and V. Narasayya, “An overview of business intelligence technology,” Communications of the ACM, vol. 54, no. 8, pp. 88–98, 2011.

[18] S. Negash, “Business intelligence,” Communications of the Association for Information Systems, vol. 13, no. 1, pp. 177–195, 2004.

[19] S. Few and P. Edge, “Dashboard design: Beyond meters, gauges, and traffic lights,” Business Intelligence Journal, vol. 11, no. 1, pp. 20–29, 2006.

[20] B. H. Wixom and H. J. Watson, “The BI-based organization,” International Journal of Business Intelligence Research, vol. 1, no. 1, pp. 13–28, 2010.

PHINMA UNIVERSITY OF ILOILO
College Of Information Technology Education

CHAPTER IV

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.1 Introduction

This chapter presents the development, implementation, testing,and evaluation of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo**. The discussion follows the structure of the Chapter 4 guide while replacing the guide's sample content with the system, processes, technologies, user roles, and database structure documented in Chapters 1–3. The developed system responds to the operational situation of Luna's Arrozcaldo, an eight-branch food business in Iloilo City that previously relied on paper-based sales records, manual inventory monitoring, and physical branch visits for consolidated management information.

The chapter explains how information moves from branch-level sales processing into a centralized PostgreSQL database and then into management dashboards and reports. It also discusses the major modules, database relationships, security controls, deployment arrangement, and testing procedures. Screenshot and diagram locations are provided as placeholders because actual screenshots must come from the final system and should not be fabricated.

The evaluation discussion follows the four criteria established in Chapter 3: functionality, usability, security, and efficiency. The formal respondents were the business owner or administrator, branch managers, and selected cashiers or counter staff. The completed Chapters 1–3 document the evaluation instrument and its administration, but they do not provide the 1 final item-level weighted means; therefore, this document does not invent numerical ratings.

The team will monitor the deployed website and address technical issues when necessary. Monthly hosting expenses of $2 will be shared equally among the five members, with each contributing $0.40. These activities support continued system availability and organized database management.

4.2 Overview of the Developed System

The developed system is a web-based **Digital Sales Processing System** designed specifically for **Luna's Arrozcaldo**. Its primary purpose is to replace paper-based sales recording with a digital point-of-sale process while providing centralized management information for the business's eight branches. The system connects sales processing, inventory, customer information, promotions, and analytical reporting through a centralized PostgreSQL database hosted on Supabase.

At the branch level, the cashier uses the POS terminal to select products, enter quantities, record customer information when applicable, apply valid discounts or promotions, calculate the transaction total, and complete the sale. A completed transaction is recorded in the central database and is associated with the branch and cashier responsible for the transaction. Transaction Items preserve the individual products included in the sale. The completed transaction also causes the corresponding inventory information to be updated.

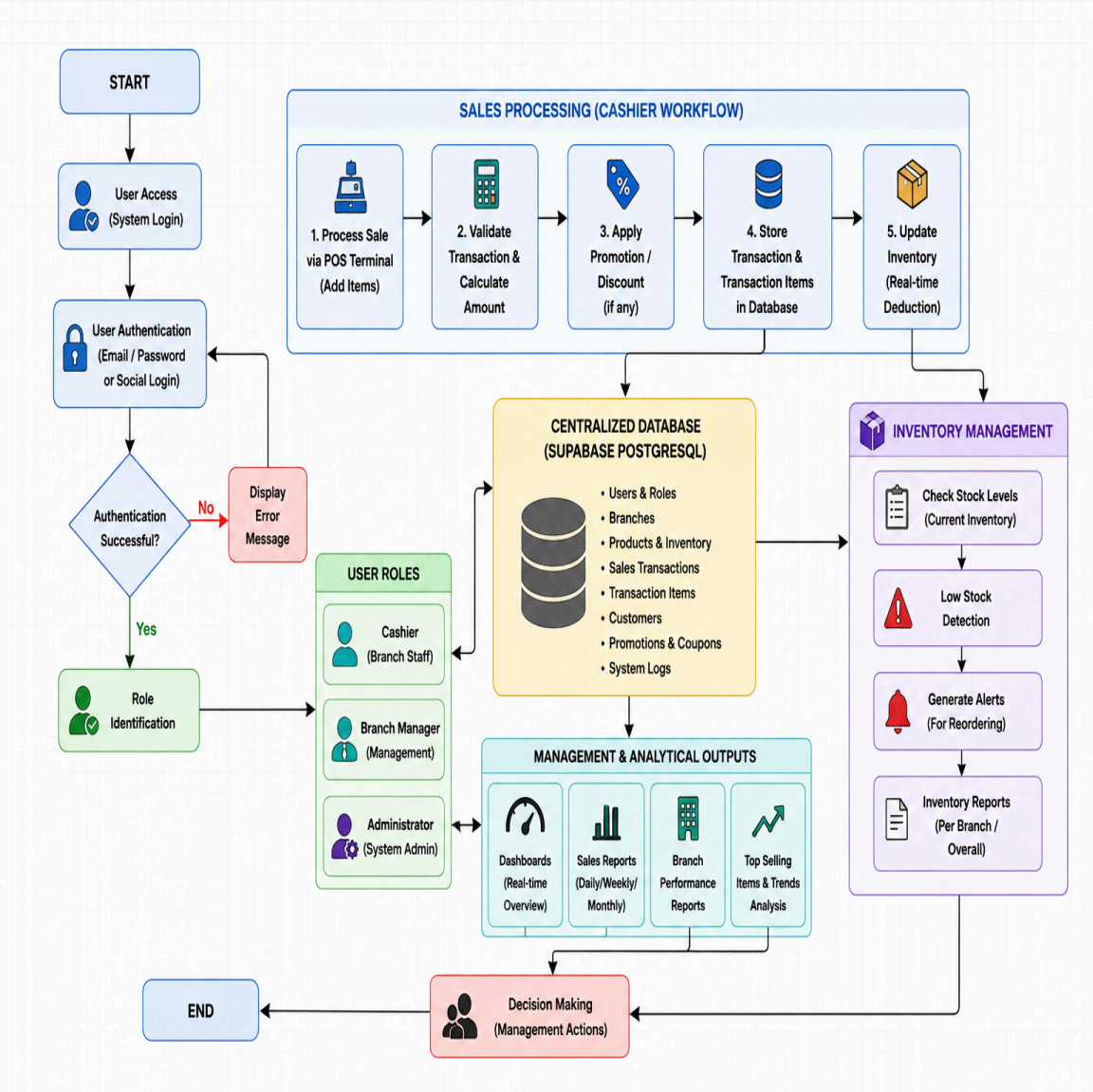
At the management level, the dashboard and reporting functions organize accumulated transaction data into information that can be reviewed without manually combining paper records. The documented reporting functions include daily, weekly, and monthly sales reports, branch performance comparisons, menu item rankings, trend analyses, inventory status, customer purchase information, and promotional information. These outputs support management decisions using a common data source.

The major objectives addressed by the system are to provide a secure web-based management dashboard, replace paper-based sales recording with a digital POS terminal, centralize information from all eight branches, generate analytical reports and visual summaries, maintain real-time inventory information, support customer profiles and purchase histories, and manage coupons and promotions. These objectives directly correspond to the problems and specific objectives stated in Chapter 1.

The developed system integrates digital sales processing, centralized sales management, and analytical reporting to help Luna’s Arrozcaldo organize business transactions across its eight branches. Its centralized PostgreSQL database stores sales and related information, allowing authorized users to retrieve records and review business performance more efficiently.

The system also connects completed sales transactions with inventory updates and reporting functions to help management monitor product demand, sales trends, and customer purchasing behavior. Accurate and consistent transaction records are important because database transactions help preserve data integrity when information is processed and updated.

Figure 4.1. Overall System Workflow Diagram



The workflow begins with user authentication and role identification. After successful authentication, the authorized user is directed to functions permitted for the assigned role. A cashier processes a sale through the POS terminal; the system validates the transaction, calculates the amount, applies an applicable promotion, stores the transaction and transaction items, and updates inventory. Management users retrieve centralized information through dashboards and reports. The connected records allow one sales event to contribute to operational records and analytical outputs.

4.3 System Development Methodology

The system was developed using the **Agile Development Model** identified in Chapter 3. Agile was selected because the project required repeated interaction with the client, progressive development of system functions, and adjustment of details based on review and feedback. Development was divided into manageable increments, with each sprint producing a working portion of the system that could be reviewed before the next increment.

The development process covered Requirements Analysis, System Design, Development, Testing, Deployment, and Evaluation. Requirements Analysis established the operational problems and system requirements through interviews, observation, and document review. System Design translated those requirements into architecture, workflow, user interface, database, and access-control structures. Development implemented the designed components, while testing examined individual and integrated behavior. Deployment prepared the application for its intended environment, followed by user acceptance and evaluation.

The development process consisted of the following phases:

Requirements Analysis

The requirements analysis phase focused on the actual sales, inventory, customer, and management practices of Luna's Arrozcaldo. The researchers identified difficulties associated with paper-based sales recording, physical branch monitoring, manual inventory checking, limited customer information, and the absence of centralized analytical reporting. These findings became requirements for digital sales processing, centralized storage, inventory management, customer management, promotions, reporting, authentication, and role-based access.

System Design

The system design phase established the architecture, data flow, entity relationships, database normalization, user roles, and interface organization. The multi-branch setting was considered by associating users, transactions, and inventory records with branches. This branch-aware structure permits centralized reporting while preserving the source location of operational records.

Development

Development transformed the approved design into a working web application. PHP was used for server-side application logic, while HTML, CSS, JavaScript, and Bootstrap were used for the web interface. PostgreSQL hosted through Supabase provides centralized data storage. Docker supports a consistent development and runtime environment, Railway is identified for production deployment, and Git supports collaborative version control.

Deployment and Evaluation

After development and testing, the system was prepared for deployment and user acceptance testing. The intended users were the business owner or administrator, branch managers, and selected cashiers or counter staff. The evaluation instrument contains twenty Likert-scale items covering functionality, usability, security, and efficiency. Chapter 3 reports twenty-five respondents and complete responses for the twenty core items, while the pilot reliability coefficients ranged from 0.78 to 0.89.

Figure 4.2. Agile Development Life Cycle



The diagram should show the iterative relationship among requirements analysis, system design, development, testing, deployment, and evaluation. Chapter 3 identifies three development sprints: Sprint 1 for POS Core, Sprint 2 for Inventory and CRM, and Sprint 3 for Analytical Reporting.

Sprint-by-Sprint Development Discussion

Sprint 1 focused on the POS core because transaction processing is the central operational function. The increment established the basic authentication and sales workflow needed for a cashier to select products, enter quantities, calculate totals, and store completed transactions. The related user, branch, product, transaction, and transaction-item relationships were established as part of this core.

*Sprint 2 f*ocused on Inventory and Customer Relationship

Management. Inventory was connected to completed sales so that stock information could be updated when transactions were processed. The design also included low-stock thresholds and records for deliveries, spoilage, adjustments, and inter-branch transfers. Customer profiles and loyalty information were connected to transaction histories to support repeat-purchase recognition.

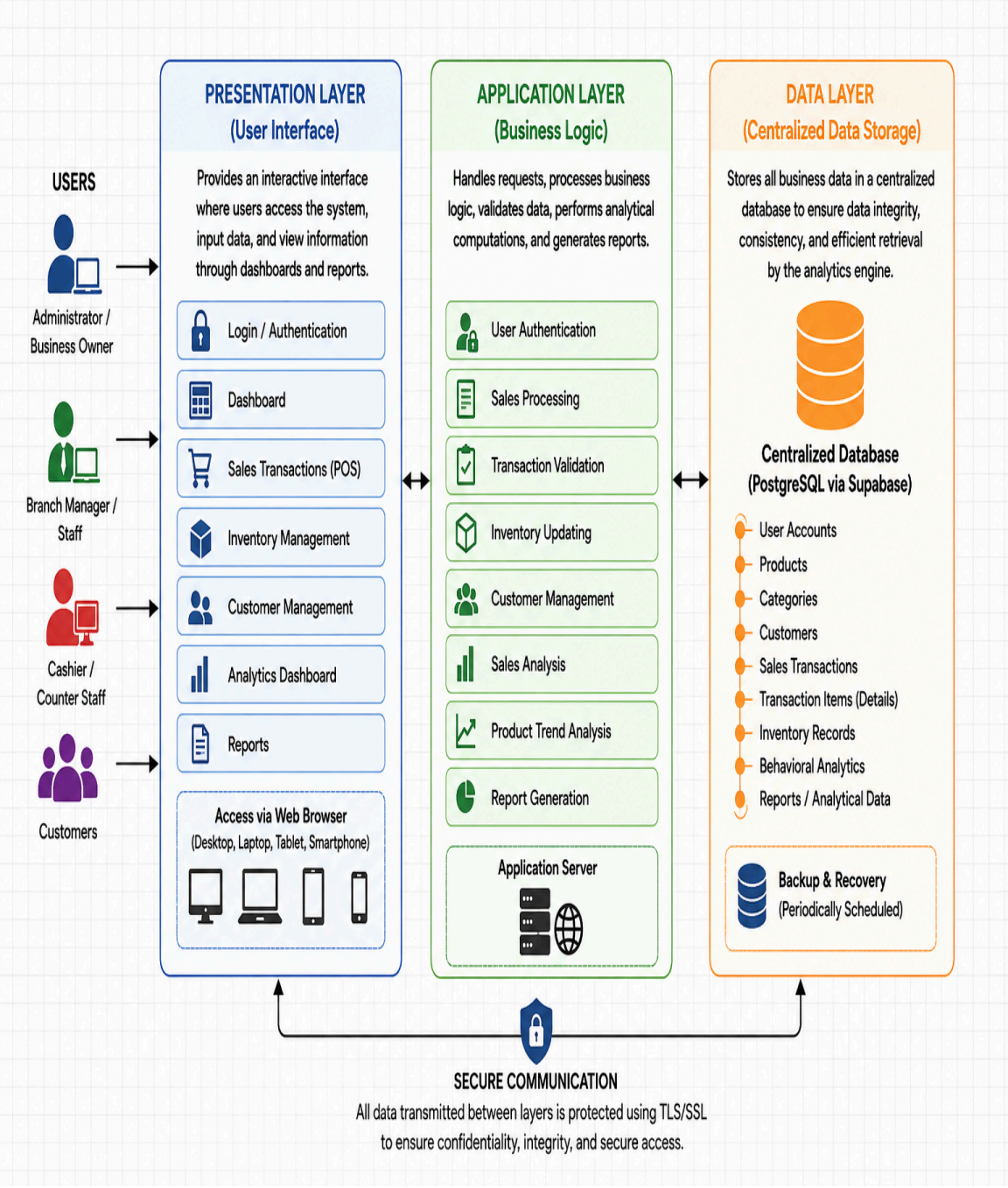
Sprint 3 focused on Analytical Reporting. Accumulated transaction data was organized into management reports and visual summaries. The documented requirements include daily, weekly, and monthly sales reports, branch comparisons, menu item rankings, and trend analyses. The dashboard was designed around management questions concerning sales performance, branch differences, inventory alerts, and active promotions.

The sprint arrangement also supported risk management. Chapter 3 identifies scheduling delays, scope creep, data loss, low respondent turnout, and connectivity problems as risks. The documented mitigation measures included early scheduling, sprint prioritization, version-controlled commits, automated Supabase backups, respondent planning, and an offline-first POS approach with synchronization on reconnection.

4.4 System Architecture

The system uses a web-based client-server arrangement organized around the user interface, application processing, and centralized data services. This separates presentation from business logic and database operations. Such separation is appropriate for a multi-branch system because branch users access the application through a browser while the centralized database remains the common repository for sales, inventory, customer, staff, and promotional information.

Figure 4.3. System Architecture Diagram



The architecture should show users accessing the web interface through a modern browser, PHP handling application logic, PostgreSQL hosted on Supabase providing centralized storage, and Railway providing the documented production deployment environment.

The architecture consists of the following components:

4.4.1 Presentation Layer

The presentation layer is the part of the system directly used by personnel. It presents login controls, dashboards, POS functions, inventory information, customer information,

promotion functions, and analytical reports according to the user's role. HTML, CSS, JavaScript, and Bootstrap support the interface. Bootstrap 5 provides consistent interface components and responsive presentation.

4.4.2 Application Layer

The application layer contains the business logic that processes user requests. PHP handles authentication and session management, validates submitted information, processes sales, calculates totals, communicates with the database, updates inventory, and prepares information for reporting. Role-based rules are applied so that users receive access appropriate to their responsibilities.

4.4.3 Data Layer

The data layer is provided by PostgreSQL hosted through Supabase. It stores branches, users, products, categories, transactions, transaction items, inventory, customers, loyalty points, promotions, and audit logs. Foreign-key constraints and indexes support data integrity and reporting performance.

4.4.4 Security and Data Integrity Layer

Security is applied across the architecture. Data transmitted through the system is protected using TLS. Passwords are stored as salted cryptographic hashes using PHP's bcrypt implementation. PostgreSQL transaction atomicity protects the consistency of POS operations. Audit logs associate significant events with user and timestamp information, providing accountability.

4.5 Hardware, Software, and Network Requirements

The system is web-based, so branch workstations need a device capable of running a modern browser and maintaining a reliable connection to the deployed application. The POS is intended

to run on a desktop or laptop at each branch counter, while authorized management users can access the web-based dashboard from an appropriate internet-connected device.

4.5.1 Hardware Requirements

Table 4.1. Hardware Requirements

Component	Practical Requirement	Purpose
Computer	Desktop or laptop capable of running a modern browser	Branch POS and management access
Memory	Sufficient RAM for browser and web application use	Normal system operation
Display	Monitor suitable for reading POS and dashboards	Transaction and management viewing
Network equipment	Reliable branch network and internet connection	Connection to centralized services
Peripherals	Standard keyboard and pointing device; compatible output hardware when used	Supports transaction processing

4.5.2 Software Requirements

Table 4.2. Software Requirements

Technology	Purpose
PHP	Backend application logic, authentication, transaction processing, database queries, and reports
PostgreSQL	Relational database management
Supabase	Cloud hosting for PostgreSQL and authentication services
HTML/CSS/JavaScript	Web interface structure, styling, and client-side behavior
Bootstrap 5	Responsive and consistent interface components
Railway	Production application deployment environment identified in Chapter 3
Git	Version control and collaborative development
Modern web browser	Access to the web-based system

4.5.1 Network Requirements

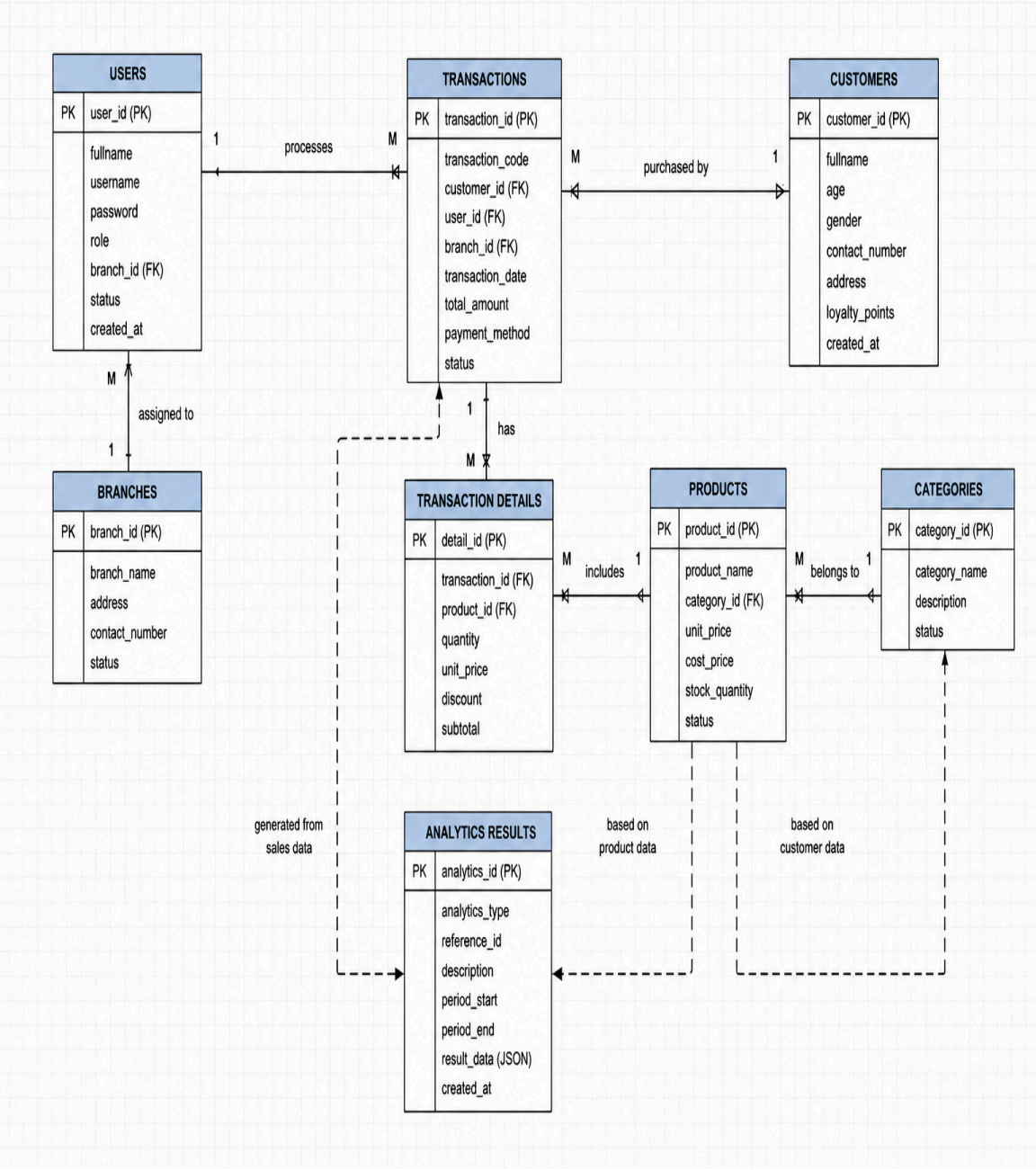
A reliable internet connection is important because the system uses centralized cloud services. The network permits branch POS terminals to communicate with the application and database and allows authorized management users to access consolidated information remotely. Chapter 3 identifies connectivity as a risk and documents an offline-first POS with synchronization on reconnection as a mitigation approach.

The system does not require a separate database at every branch. Branches access the common web application and centralized database. This reduces duplicate data stores and makes the centralized database the authoritative source of operational information. Secure communication is required whenever information is transmitted between users, the application, and cloud services.

4.6 Database Design

The database serves as the central storage foundation of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting** by organizing sales transactions, customer information, product records, and analytical data. The system uses a **relational database structure** to reduce data redundancy, maintain data integrity, and support efficient retrieval of information required for centralized sales management and analytical reporting. The database design was derived from the system requirements and the relationships among users, customers, products, transactions, transaction details, and analytics.

Figure 4.4. Entity Relationship Diagram



The Entity Relationship Diagram **(ERD)** presents the relationships among the major entities that support the system's sales processing, customer management, product management, transaction recording, and analytical reporting functions. The primary entities of the database include **Users, Customers, Products, Transactions, Transaction Details, and Analytics**, which work together to maintain organized operational and analytical records.

The primary entities of the database include:

- **Users** – stores the accounts and role information of authorized system users.
- **Customers** – maintains customer information used to support customer management and behavioral analysis.
- **Products** – stores menu products and their corresponding product information.
- **Transactions** – records completed sales processed through the digital sales processing system.
- **Transaction Details** – records the individual products included in each completed transaction.
- **Analytics** – stores or supports processed analytical information derived from historical transaction data for reporting and decision-making.

Users Table

The Users Table stores the account information of authorized personnel who access and operate the system. It supports user authentication and role-based access by identifying the user and the level of access appropriate to their responsibilities.

Stores account information, including user identification, login credentials, assigned roles, branch association, and account status, as represented in the database design. It supports authentication and role-based access by helping the system identify authorized personnel and determine their permitted activities. Primary keys and appropriate access restrictions help maintain reliable user records and protect sensitive account information

Field	Description
user_id	Primary Key
full_name	Complete name of the user
username	Login username
password	Stored password information
role	User access level or assigned responsibility

Customers Table

The Customers Table stores customer information used by the system to maintain customer records and support the analysis of purchasing behavior. These records provide information that can be associated with sales transactions and used as a basis for customer-related analytical reporting.

Maintains customer information that can be associated with sales transactions and used to understand purchasing behavior. Linking customer records to transactions allows the system to organize purchase histories and support customer-related analytical reporting. Appropriate data validation and access restrictions help maintain accurate customer records while limiting unnecessary exposure of personal information.

Field Description

customer_id Unique identifier of the customer

fullname Customer's complete name

age Customer age

gender Customer gender

Products Table

The Products Table contains the products or menu items managed by the system. It provides the product information needed during sales processing and allows transaction records to identify which products were purchased.

Stores product details, including product identifiers, names, categories, selling prices, and availability status, according to the fields shown in the database design. Its relationship with transaction details allows the system to identify which products were purchased and analyze product demand. Proper constraints and consistent product records help reduce invalid entries and support reliable sales calculations and inventory monitoring.

Field Description

product_id Unique identifier of the product

category_id Identifier of the product category

product_name Name of the product or menu item

selling_price Selling price of the product

active_status Indicates whether the product is currently active

date_added Date the product was added to the system

Transactions Table

The Transactions Table records completed sales processed through the system. It serves as the main record of each sales transaction and connects the transaction with the customer and the user responsible for processing it when applicable.

Records sales information, including transaction identification, customer association when applicable, transaction date, total amount, payment method, and status, as defined by the documented schema. It serves as the main record of a sale and connects the responsible user and related transaction details to the corresponding business activity. Maintaining valid relationships between transactions and their associated records helps preserve sales history and supports accurate reporting.

Field Description

transaction_id Unique identifier of the transaction

user_id Identifies the user who processed the transaction

customer_id Identifies the associated customer, when

applicable

transaction_date Date and time the transaction was recorded

total_amount Total amount of the completed transaction

Transaction Details Table

The Transaction Details Table records the individual products included in each sales transaction. It provides the detailed connection between transactions and products, allowing the system to determine the items purchased, their quantities, and their corresponding amounts.

Records the individual products included in each transaction, together with their quantities, prices, discounts, and subtotals where those fields are implemented. Its relationship with the Transactions and Products tables enables the system to organize multi-item sales without repeating the complete transaction record for every product. Valid foreign keys and consistent calculation rules help maintain the accuracy of item-level records and provide a dependable basis for sales analysis.

Field	Description
transaction_detail_id	Unique identifier of the transaction detail
transaction_id	Identifies the related transaction
product_id	Identifies the purchased product
quantity	Number of units purchased
price	Price of the product at the time of the transaction
subtotal	Calculated amount for the product based on quantity

Analytics Table

The Analytics Table supports the system's analytical reporting by organizing information derived from historical sales and customer transaction records. The stored analytical information can be used to identify purchasing patterns, frequently purchased products, and sales trends that support management in making data-driven decisions.

As represented in the provided ERD, supports the organization of analytical results derived from sales, product, and customer information. Its documented fields, such as analysis type, reference identifiers, period dates, and result data, can help organize outputs used for management reporting and decision-making. The actual implementation should be checked to confirm which results are stored in this table and which are calculated dynamically from the related records.

Field	Description

analytics_id Unique identifier of the analytical record

transaction_id Identifies the transaction used as an analytical reference

product_id Identifies the product included in the analysis

customer_id Identifies the customer associated with the

analytical record, when applicable

analysis_type Identifies the type or category of analysis

analysis_result Stores the resulting analytical information

The relationships among the tables enable the system to track customer purchases, recognize repeated buying patterns, determine product demand, and produce comprehensive analytical reports. Proper indexing and foreign key constraints maintain data integrity and improve query performance, allowing the system to efficiently handle both sales transactions and analytical processes.

The relationships among the Users, Customers, Products, Transactions, Transaction Details, and Analytics tables enable the system to organize sales records, track customer purchases, identify product demand, and support analytical reporting. Primary keys and foreign key constraints help maintain relationships between records, while appropriate indexing can improve the efficiency of database queries. These relationships provide a structured foundation for centralized sales management, accurate transaction processing, and data-informed decision-making for Luna's Arrozcaldo

---

4.7 Module Descriptions

The system is organized into functional modules that correspond to the business processes identified in Chapter 1 and the use cases described in Chapter 3. The modules share the centralized database, allowing a completed sale to affect transaction records, inventory, customer history, loyalty information, promotion records, and analytical outputs. Role-based access controls which functions each user can perform.

4.7.1 Administrator Module

The Administrator module is intended for the business owner or administrator. It supports system-wide oversight and management of users, branches, products, promotions, and centralized business information. Administrative actions are subject to role-based access rules and significant changes can be associated with the responsible account through audit information.

4.7.2 Branch Manager Module

The Branch Manager module supports day-to-day branch oversight. A branch manager can review branch-related sales and inventory information and use reports to understand the performance of the assigned location. The branch-aware database preserves the connection between operational records and their source branch.

4.7.3 Cashier Module

The Cashier module centers on the POS terminal. The cashier selects products, enters quantities, records customer information when applicable, applies valid discounts or promotions, confirms the total, and completes the transaction. The completed transaction is stored centrally and inventory is updated. This replaces the paper-based sales process.

4.7.4 Inventory Module

The Inventory module provides branch-level stock visibility. It keeps inventory information current as sales are processed and maintains records of deliveries, spoilage, adjustments, and inter-branch transfers. It also provides low-stock alerts when configurable thresholds are crossed.

4.7.5 Sales Processing Module

Sales Processing is the core operational function. The cashier selects products and quantities, reviews the order, and completes the transaction. The application retrieves prices, calculates totals, applies an applicable promotion, and records the sale. The transaction is associated with its branch and cashier and may be associated with a customer.

4.7.6 Customer Management Module

Customer Management supports digital customer profiles and purchase histories. The information provides a basis for identifying repeat customers and managing loyalty points. Personal information is handled in accordance with the Data Privacy Act of 2012 as stated in the study.

4.7.7 Promotions and Coupon Management Module

The Promotions module enables management to create digital discount offers, define their discount type and value, set

validity periods, and determine branch scope. A valid promotion 18 can be associated with a transaction during sales processing, creating a record that can be considered in promotional reporting.

4.7.8 Reporting and Analytics Module

The Reporting and Analytics module transforms stored transaction data into management information. Documented outputs include daily, weekly, and monthly sales reports, branch comparisons, menu item rankings, trend analyses, inventory status, customer purchase behavior, and revenue trends. Visual dashboards make these outputs easier to interpret.

4.7.9 Role-Based Access Control

Role-based access limits functions according to responsibility. The three principal roles are administrator, branch manager, and cashier. The administrator has the broadest management authority, branch managers work primarily with branch operations and management information, and cashiers focus on transaction processing. Authentication verifies credentials before protected functions become available.

Maintenance includes monitoring the system, correcting identified issues, reviewing logs, maintaining backups, and applying necessary updates. Agile also supports maintenance because user feedback can become a future iteration. Changes should be prioritized so that improvements do not compromise sales records, inventory data, or reporting.

CHAPTER V

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusion of the development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo. It brings together the problems identified in Chapter 1, the methodology and technical decisions in Chapter 3, and the implementation and testing discussion in Chapter 4.

The development of the Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo focuses on improving the accuracy, accessibility, and organization of business records. By integrating sales transactions, inventory monitoring, and reporting, the system supports more efficient daily operations and provides management with useful information for decision-making. Similar POS research highlights how integrating transaction processing, inventory tracking, and report generation can help reduce manual recording errors and improve access to sales information

---

5.2 Summary of Findings

The development and evaluation of the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo** produced findings aligned with the study objectives:

- The system provided centralized sales management by connecting transactions from Luna's Arrozcaldo's eight branches to a shared database, allowing authorized management users to access consolidated information.
- The digital POS improved sales processing by organizing product selection, quantity entry, total calculation, promotion application, and transaction recording, reducing manual recording steps.

- The centralized sales data supported analytical reporting through daily, weekly, and monthly reports, branch comparisons, product rankings, and sales trend analysis for better decision-making.
- The system integrated inventory management with sales transactions, allowing completed sales to update stock records and supporting low-stock monitoring and other inventory movements.
- Customer and promotion features supported customer management by connecting customer profiles with purchase histories and loyalty points while allowing promotions to include discounts, validity dates, and branch coverage.
- Security and accountability were strengthened through authentication, role-based access, encrypted passwords, TLS, audit logs, database constraints, and automated backups.
- The Agile development approach supported continuous improvement through incremental development of the POS, inventory and customer management, and analytical reporting features, with review and adjustments during development.
- The system was evaluated based on functionality, usability,security, and efficiency, with respondent administration and evaluation procedures documented as part of the system assessment.
- Analytical Reporting: Organized sales data can be used to generate daily, weekly, and monthly summaries, compare branch performance, and identify product sales patterns. These reports provide management with information to support business planning and decision-making.
- Security and Accountability: Authentication, role-based access, secure database queries, and appropriate activity logging help protect business information from unauthorized access. Applying these practices supports the confidentiality and integrity of sales and customer records.
- Centralized Information Management: The system brings sales records from different branches into a shared database, helping authorized managers review consolidated information. This supports easier monitoring and coordination across Luna's Arrozcaldo branches.
- Improved Transaction Processing: Digital product selection, quantity entry, total calculation, and transaction recording help establish a consistent sales workflow. These functions can reduce repetitive manual work and recording errors.
- Inventory Monitoring: Connecting sales transactions with product records helps the business monitor stock movement and identify items that may need replenishment. Accurate inventory information supports better product availability and stock management.

5.3 Conclusion

The Digital Sales Processing System with Centralized Sales Management and Analytical Reporting provides a practical digital response to the operational problems identified at Luna's Arrozcaldo. The system moves the organization from fragmented paper-based recording toward a centralized information environment in which transactions, inventory, customer information, promotions, and management reports are connected.

The system addresses the need for improved visibility by allowing authorized management users to access information through a web-based dashboard rather than relying solely on physical collection of branch records. Its branch-aware database preserves the origin of transactions and inventory while allowing information to be consolidated.

The digital POS addresses weaknesses in manual sales recording through a consistent transaction workflow. The system calculates totals, supports promotions, stores transaction details, and connects completed sales with inventory and reporting. This creates more structured and traceable transaction records.

The reporting and analytics component transforms transaction records into organized management information. Daily, weekly, and monthly reports, branch comparisons, menu rankings, and trend analyses allow management to examine business activity from different perspectives.

Inventory, customer, and promotion functions extend the system beyond basic transaction recording. Inventory monitoring supports current stock information and low-stock awareness. Customer profiles and loyalty information provide a basis for relationship management. Promotions provide a structured method for defining discount campaigns.

Security is a necessary part of the conclusion because the system centralizes business records. Authentication, role-based access, bcrypt password hashing, TLS, audit logs, relational integrity constraints, and database backups provide complementary controls for confidentiality, integrity, accountability, and recovery.

Overall, the project demonstrates how digital technology can be tailored to a heritage food business without changing its established service identity. The system applies technology primarily to internal sales, inventory, customer, promotional, and management processes, strengthening the information infrastructure behind Luna's Arrozcaldo.

5.4 Recommendations

To further improve the **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo**, the following recommendations are proposed based on the system's current functions, findings, and identified future needs:

System Improvements

- Improve inventory monitoring by adding clearer low-stock alerts to help staff identify products that need replenishment on time.

- Enhance analytical reporting by providing more detailed sales summaries, branch comparisons, product rankings, and sales trends to support management decisions.

- Expand report export options so authorized users can easily save and share reports for business monitoring and record-keeping.

- Strengthen data security and backup procedures to protect sales, customer, inventory, and management records from unauthorized access or data loss.

Improve transaction validation by checking product quantities, prices, and required fields before saving a sale to reduce incomplete or inaccurate records [1], [2].

Strengthen database integrity by applying appropriate primary keys, foreign keys, and validation constraints to maintain consistent relationships among sales, customers, and product records [3].

Enhance branch-level monitoring by providing authorized managers with clear comparisons of sales totals and transaction activity across branches [4].

Improve the search and filtering functions so authorized users can locate transactions by date, branch, product, or transaction number more efficiently [4].

Review database query performance and apply suitable indexing where needed to improve the retrieval of sales records and analytical reports [3].

Improve error messages and system feedback so staff can identify invalid entries, understand failed operations, and correct mistakes without unnecessary repetition [1], [2].

Future Enhancements

● Develop sales forecasting features that can use accumulated historical sales data to help predict future sales and inventory needs.

● Explore a mobile-friendly management interface to allow authorized managers to conveniently monitor sales and reports using mobile devices.

● Consider additional analytical features that can provide deeper insights into customer purchasing behavior, product performance, and branch sales trends.

Develop sales forecasting features using historical transaction data to estimate future sales and support inventory planning.

Explore automated reorder recommendations based on product sales trends and recorded stock levels to help staff plan replenishment.

Consider adding customizable dashboards that allow managers to select reporting periods, branches, and product categories for more focused analysis.

Explore customer purchasing trend analysis to help management understand recurring purchases and plan suitable promotions while protecting customer information.

Consider developing a responsive mobile interface so authorized managers can review sales summaries and reports on different screen sizes.

Explore integrating notification features for important events, such as low-stock conditions, unusual transaction activity, or unsuccessful backup operations.

Evaluate the possibility of integrating additional business tools in the future, provided that the integration is secure, compatible with the existing database, and appropriate for the restaurant's needs [4], [5].

Implementation Strategies

● Provide proper training for staff and managers so they can correctly use the POS, inventory, customer management, and reporting functions.

● Conduct regular system maintenance and data backups to keep the system reliable, secure, and updated.

● Continue gathering user feedback from cashiers, branch managers, and administrators to identify areas that can be improved in future system updates.

● Maintain consistent data entry practices across all branches to ensure that centralized reports and analytics remain accurate and useful.

- Prepare a structured implementation schedule that identifies system deployment tasks, responsible personnel, target dates, and expected outcomes.
- Conduct user training and practical demonstrations for cashiers, branch managers, and administrators before introducing system changes into daily operations.
- Establish a regular backup schedule and periodically test data restoration to help ensure that sales and inventory information can be recovered after a system failure [5].
- Apply role-based permissions and review user accounts regularly so each employee can access only the information and functions required for their responsibilities [4].
- Create a maintenance checklist covering software updates, error reviews, database integrity, access permissions, and system performance to support reliable operation [3], [5].
- Collect feedback from staff and management after implementation, document recurring problems, and prioritize improvements according to their effect on transaction accuracy, usability, security, and reporting [1], [2].

5.5 Impact of the Study

The **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo** provides a practical digital solution to the operational challenges experienced by the business across its eight branches in Iloilo City. By combining digital sales processing,centralized sales management, inventory management, customer management, promotions, and analytical reporting, the system helps organize important business information in one centralized environment.

For **Luna's Arrozcaldo**, the system reduces dependence on manual sales recording and physical collection of branch information. Sales transactions recorded through the digital POS can be stored centrally and connected with inventory updates, customer purchase history, promotions, and management reports. This gives management a more organized basis for monitoring branch performance and making informed business decisions.

For **branch managers and cashiers**, the system provides a more consistent and structured way of handling sales and inventory activities. It reduces handwritten calculations and recording while making important information easier to access. The centralized system also supports continuity of

business information by providing a formal digital process instead of relying mainly on manual staff-to-staff information sharing.

The study also demonstrates how **digital technology can support a long-established heritage food business without changing its established identity and service practices.** Instead, technology is applied mainly to sales, inventory, customer, promotional, and management processes, strengthening the information foundation behind Luna's Arrozcaldo while supporting its continued operations.

The screenshots included in the project show the dashboard, point-of-sale interface, inventory page, analytics view, customer and order records, and payment-selection sequence, providing a concrete record of the interface developed by the researchers. Reviewing these pages across desktop and mobile screen sizes, with attention to readable text, clear controls, and keyboard operation, can help assess usability against the Web Content Accessibility Guidelines (WCAG) 2.2 [6]. However, the screenshots document the visible interface and should not be treated alone as proof of database persistence, access-control enforcement, or improved business performance.

For continued system maintenance, the researchers should document identified defects, review security-related code changes, retest corrected functions, and preserve version information so that the testing process can be repeated. These practices are consistent with the secure development and risk-reduction recommendations of the National Institute of Standards and Technology (NIST) Secure Software Development Framework [7]. Maintaining these records also helps distinguish safeguards that have been verified through testing from features that remain planned, incomplete, or unverified.

Because the system may store customer and staff information, the collection, access, retention, and disclosure of personal data should be handled responsibly in accordance with the Philippine Data Privacy Act of 2012 [8]. Administrator, branch manager, and cashier accounts should be tested separately to verify that each role can access only the records and operations required for its responsibilities [9]. This is important because centralized sales management should make information more accessible to authorized personnel without allowing unnecessary access to personal data or branch records.

5.6 Summary

The **Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna's Arrozcaldo** represents a direct response to the operational problems identified in the study. By connecting a digital POS, centralized database, inventory management, customer management, promotions, and analytical reporting, the project establishes a unified information environment for an eight-branch food business.

The system's value lies not only in replacing paper with software but in connecting individual business activities into a common flow of information. A sale recorded at the POS can become part of the transaction history, inventory update, customer purchase history, promotion record, and management report. This connection provides the foundation for more organized operations and informed management.

The project also demonstrates the importance of developing technology around actual user requirements. Agile iterations, client review, role-based access, a normalized database, security controls, and user acceptance evaluation all support a system built around practical needs. Future improvements should continue to respond to verified operational requirements and user feedback.

In conclusion, the developed system provides Luna’s Arrozcaldo with a structured digital foundation for modern sales processing and centralized management while remaining aligned with its established business operations in Iloilo City.

This study presents SUGILANON as a web-based Digital Sales Processing System with Centralized Sales Management and Analytical Reporting for Luna’s Arrozcaldo. The project focuses on connecting sales transactions, product records, inventory information, customer details, and management reports within an organized digital environment. The submitted screenshots document important interface components, including the dashboard, point-of-sale screen, inventory page, analytics view, and customer order records. Nevertheless, the visible interface alone does not establish that every function operates correctly under actual business conditions or that measurable improvements have already been achieved.

A central purpose of the system is to organize sales transactions so that individual purchases can be reviewed and used for management reporting. Each completed transaction should be traceable through its transaction identifier, item details, quantities, prices, payment information, and applicable branch or user records. When a single sales operation updates several related database records, testing should verify that successful operations remain consistent and that failed operations do not leave incomplete or conflicting information, following the principles of database transaction processing [10]. Test cases should therefore cover valid purchases, incorrect quantities, discount calculations, payment selection, repeated submissions, transaction retrieval, and inventory changes where applicable.

The analytical reporting component is intended to help management examine daily, weekly, and monthly sales information, identify product performance, compare branches, and review changes

over time. To make these reports transparent, each output should identify the reporting period, branches included, transaction statuses counted, and rules used to calculate its totals. Another authorized tester should be able to repeat the same report using the same dataset and filters and obtain consistent results. Any comparison between manual and digital processing should also document the sample transactions, testing conditions, completion times, calculation errors, and excluded records before conclusions about efficiency or accuracy are made.

The system's usefulness also depends on responsible data handling and appropriate access permissions for the administrator, branch manager, and cashier. Authentication and authorization should be evaluated separately because verifying a user's identity does not automatically mean that the user should have unrestricted access to every system function [9]. Security reviews, documented defects, and repeated testing after corrections can help the development team maintain the system more responsibly as changes are introduced [7]. These procedures provide a stronger basis for evaluating the confidentiality, integrity, and reliability of sales and customer information without claiming that security has been guaranteed solely by implementing a web-based platform.

Overall, SUGILANON provides a structured digital foundation intended to support sales processing, centralized information management, and analytical reporting for Luna's Arrozcaldo. Its practical value should be assessed through documented functional tests, report reconciliation, usability feedback, and recorded evidence from the intended operating environment. For reproducibility, the researchers should preserve the test procedures, test data with personal information removed, expected results, actual results, screenshots, defect records, and evaluation instruments used during assessment. With these records and clearly stated limitations, the capstone can present a more credible account of the system's capabilities while identifying improvements that can be pursued in future development.

REFERENCES

[21] F. D. Davis, "Perceived usefulness, perceived ease of use, and user acceptance of information technology," MIS Quarterly, vol. 13, no. 3, pp. 319–340, 1989.

[22] J. Highsmith and A. Cockburn, "Agile software development: The business of innovation," Computer, vol. 34, no. 9, pp. 120–127, 2001.

[23] S. Negash, "Business intelligence," Communications of the Association for Information Systems, vol. 13, no. 1, pp. 177–195, 2004.

[24] E. F. Tupas and R. C. Lorente, “Heritage food establishments and digital transition in Philippine regional cities: Challenges and opportunities,” Journal of Philippine Development, vol. 45, no. 2, pp. 23–47, 2018.

[25] G. Westerman, D. Bonnet, and A. McAfee, Leading Digital: Turning Technology into Business Transformation. Harvard Business Review Press, 2014.

[26] B. H. Wixom and H. J. Watson, “The BI-based organization,” International Journal of Business Intelligence Research, vol. 1, no. 1, pp. 13–28, 2010.

[27] M. G. T. Gañgan, “Web-based market goods trading system of Agri-Pinoy Trading Center,” International Journal of Multidisciplinary Research and Arts, vol. 8, no. 6, 2025.

[6] World Wide Web Consortium (W3C), “Web Content Accessibility Guidelines (WCAG) 2.2,” W3C Recommendation, Dec. 12, 2024.
Web Content Accessibility Guidelines (WCAG) 2.2.

[7] K. Scarfone, M. Souppaya, and D. Dodson, “Secure Software Development Framework (SSDF) Version 1.1: Recommendations for Mitigating the Risk of Software Vulnerabilities,” NIST Special Publication 800-218, Feb. 2022.
https://csrc.nist.gov/pubs/sp/800/218/final

[8] Republic of the Philippines, “Republic Act No. 10173: Data Privacy Act of 2012,” National Privacy Commission.
https://privacy.gov.ph/data-privacy-act-/.

[9] OWASP Foundation, “Authorization Cheat Sheet,” OWASP Cheat Sheet Series.: https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html

[10] PostgreSQL Global Development Group, “Transaction Processing,” PostgreSQL Documentation.
https://www.postgresql.org/docs/current/transactions.html.